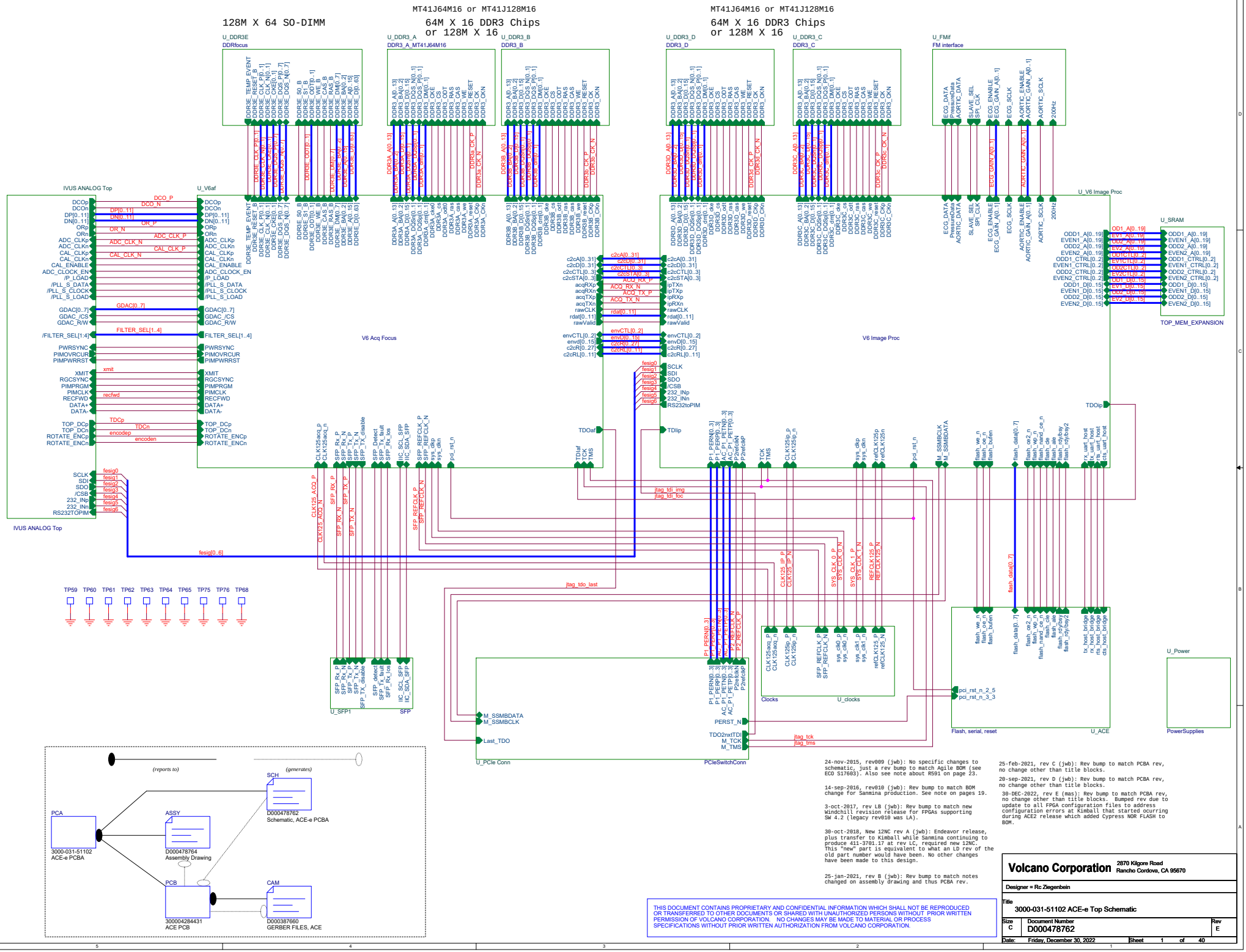


24-nov-2015, rev009 (jwb): No specific changes to schematic, just a rev bump to match Agile BOM (see ECO S17603). Also see note about R591 on page 23.

14-sep-2016, rev010 (jwb): Rev bump to match BOM change for Sanmina production. See note on pages 19.

3-oct-2017, rev LB (jwb): Rev bump to match new Windchill revision release for FPGAs supporting SW 4.2 (legacy rev010 was LA).

30-oct-2018, New 12NC rev A (jwb): Endeavor release, plus transfer to Kimball while Sanmina continuing to produce 411-3701.17 at rev LC, required new 12NC. This "new" part is equivalent to what an LD rev of the old part number would have been. No other changes have been made to this design.

25-jan-2021, rev B (jwb): Rev bump to match notes changed on assembly drawing and thus PCBA rev.

25-feb-2021, rev C (jwb): Rev bump to match PCBA rev, no change other than title blocks.

20-sep-2021, rev D (jwb): Rev bump to match PCBA rev, no change other than title blocks.

30-DEC-2022, rev E (mas): Rev bump to match PCBA rev, no change other than title blocks. Bumped rev due to update to all FPGA configuration files to address configuration errors at Kimball that started ocurring during ACE2 release which added Cypress NOR FLASH to BOM.

THIS DOCUMENT CONTAINS PROPRIETARY AND CONFIDENTIAL INFORMATION WHICH SHALL NOT BE REPRODUCED OR TRANSFERRED TO OTHER DOCUMENTS OR SHARED WITH UNAUTHORIZED PERSONS WITHOUT PRIOR WRITTEN PERMISSION OF VOLCANO CORPORATION. NO CHANGES MAY BE MADE TO MATERIAL OR PROCESS SPECIFICATIONS WITHOUT PRIOR WRITTEN AUTHORIZATION FROM VOLCANO CORPORATION.

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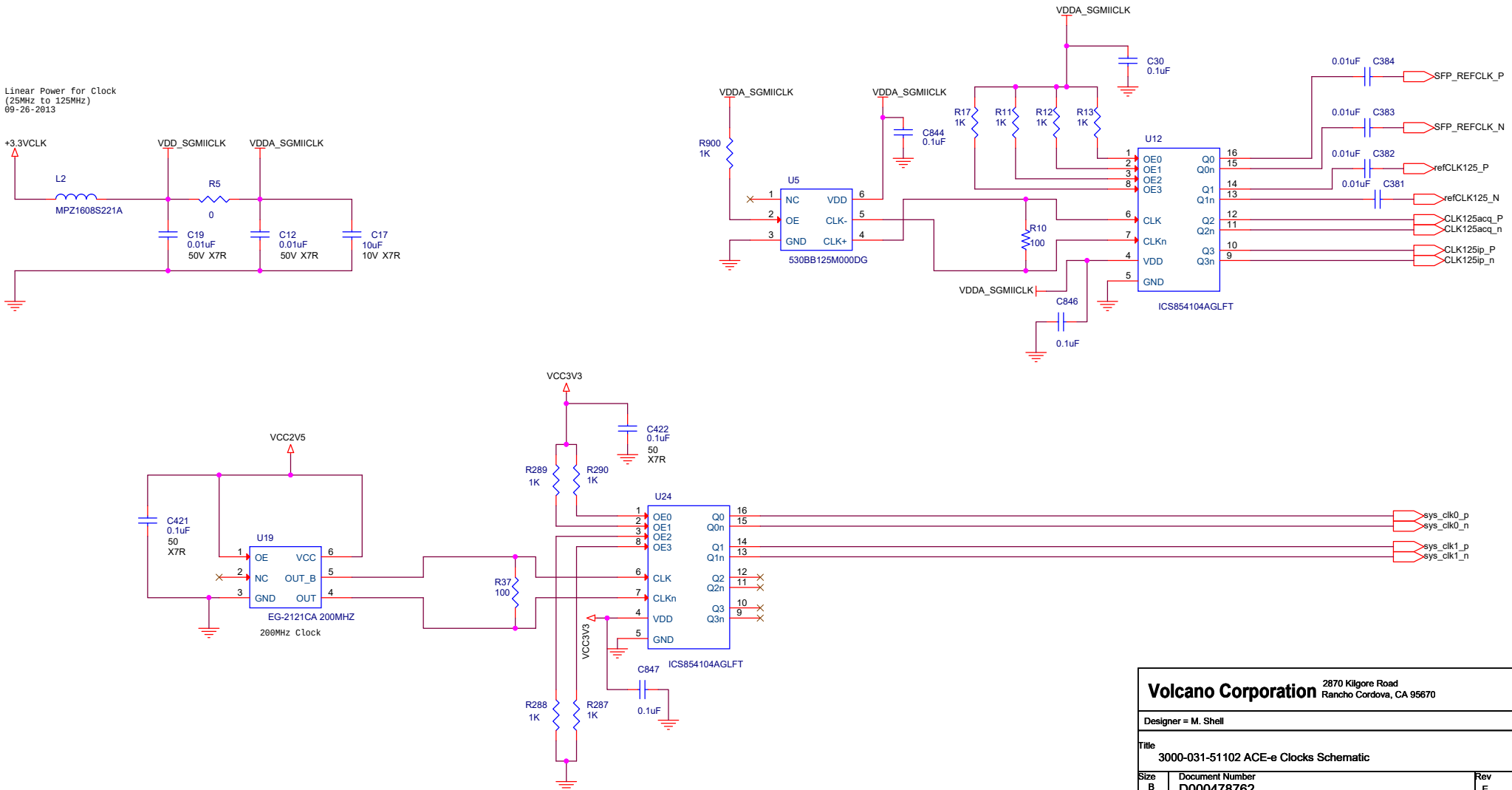
Designer = Rc Ziegenbein

Title: 3000-031-51102 ACE-e Top Schematic

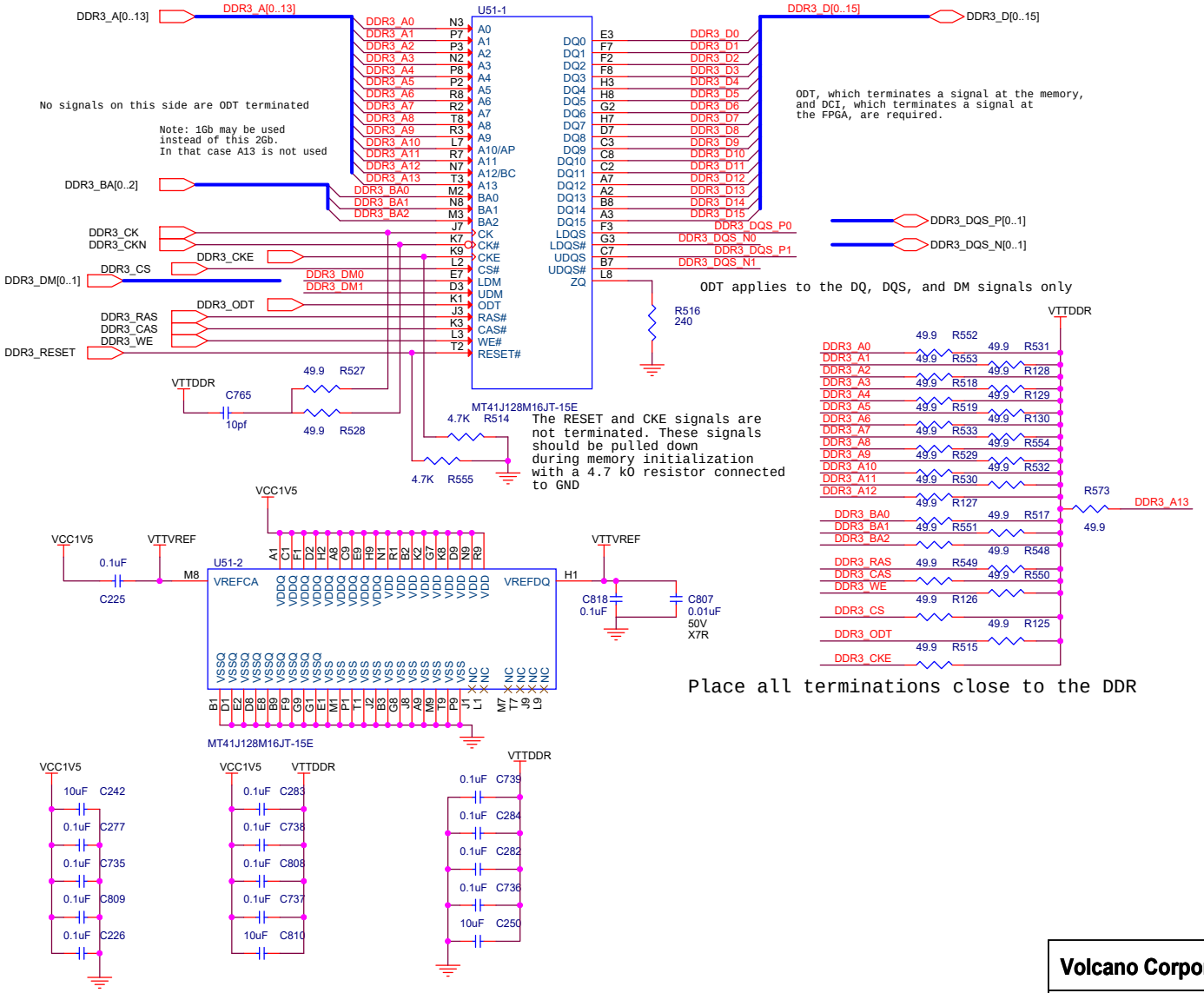
Size	Document Number	Rev
C	D000478762	E

Date: Friday, December 30, 2022 Sheet 1 of 40

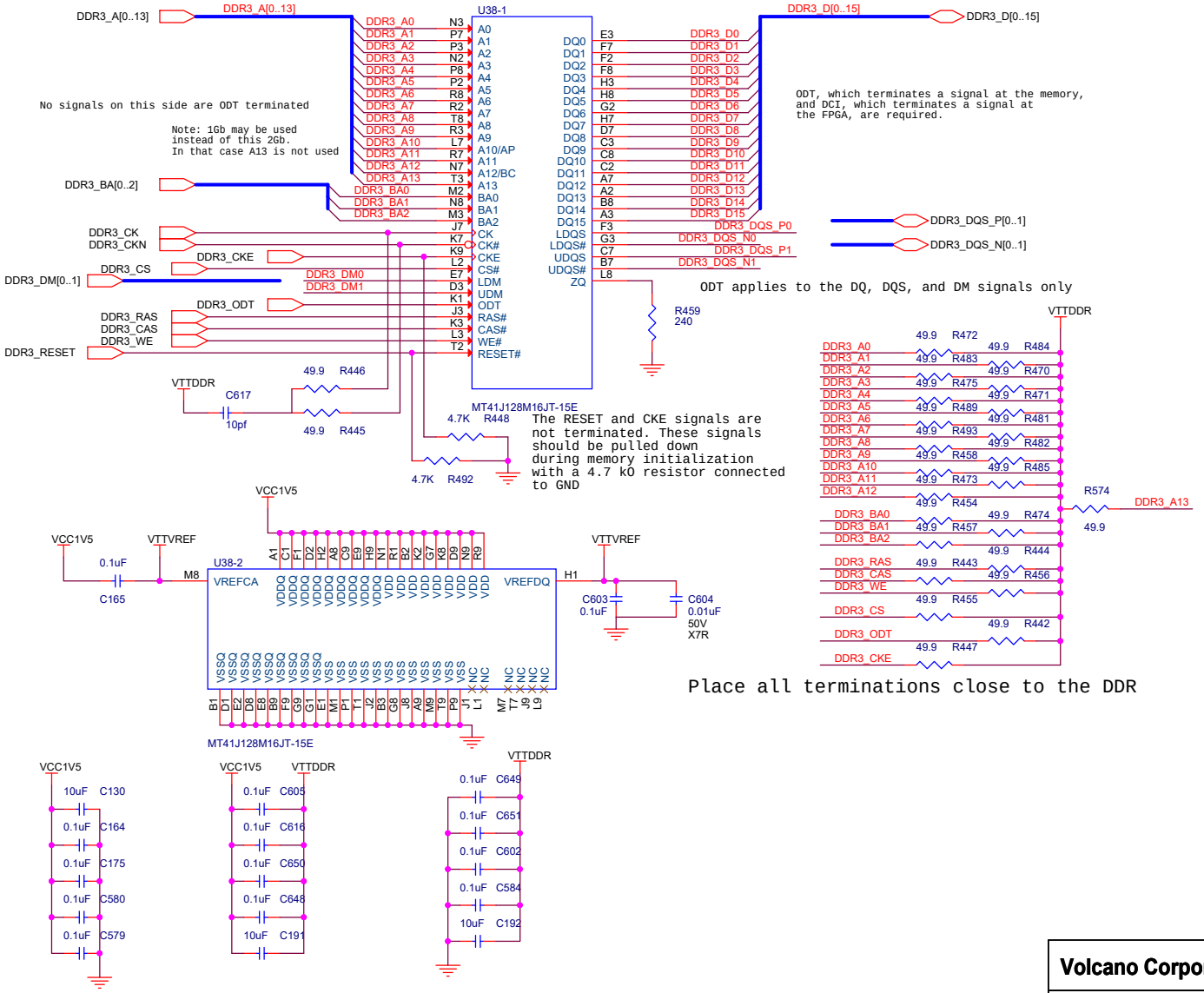
Linear Power for Clock
(25MHz to 125MHz)
09-26-2013



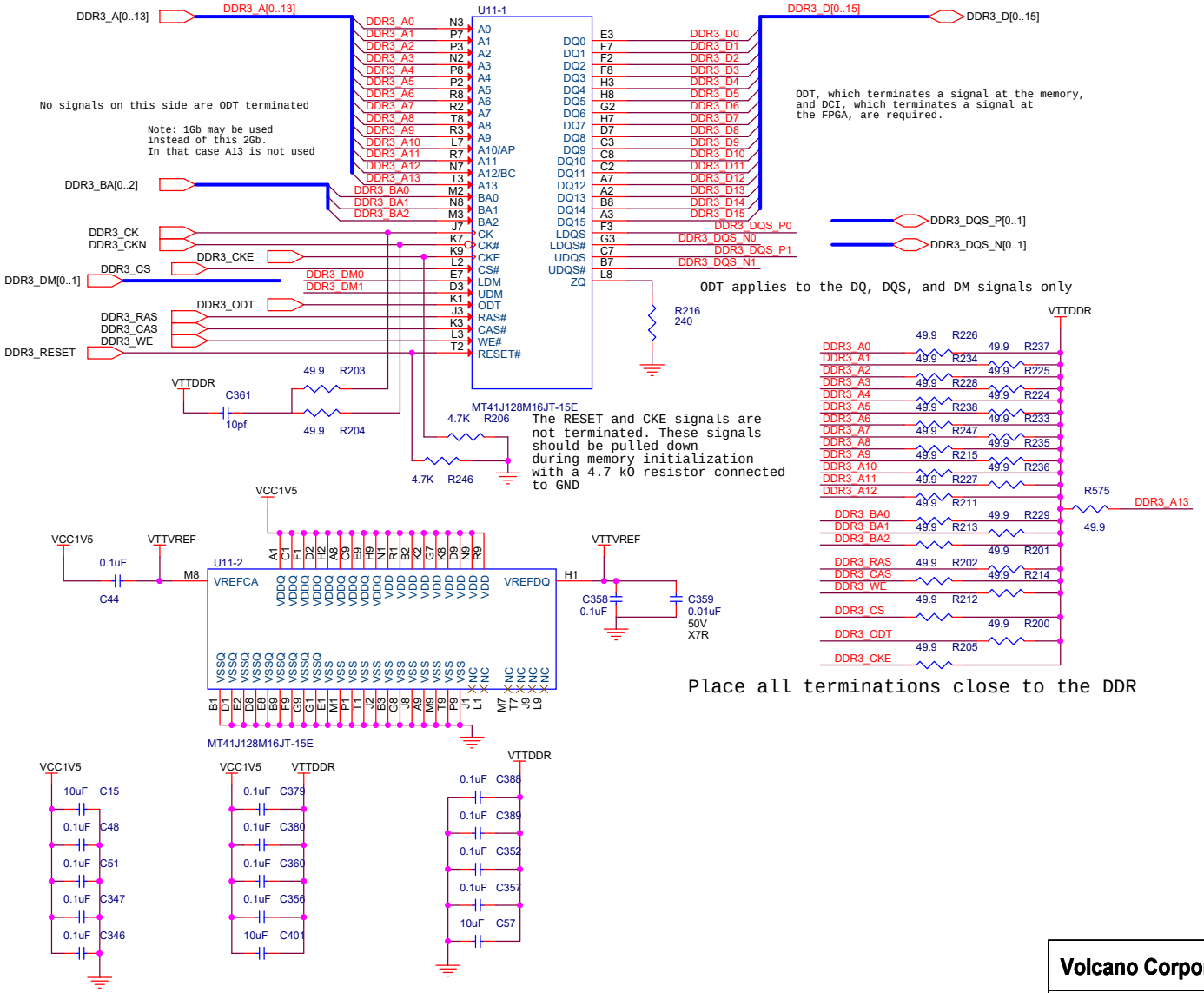
Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = M. Shell		
Title 3000-031-51102 ACE-e Clocks Schematic		
Size B	Document Number D000478762	Rev E
Date: Friday, December 30, 2022	Sheet 2 of 40	



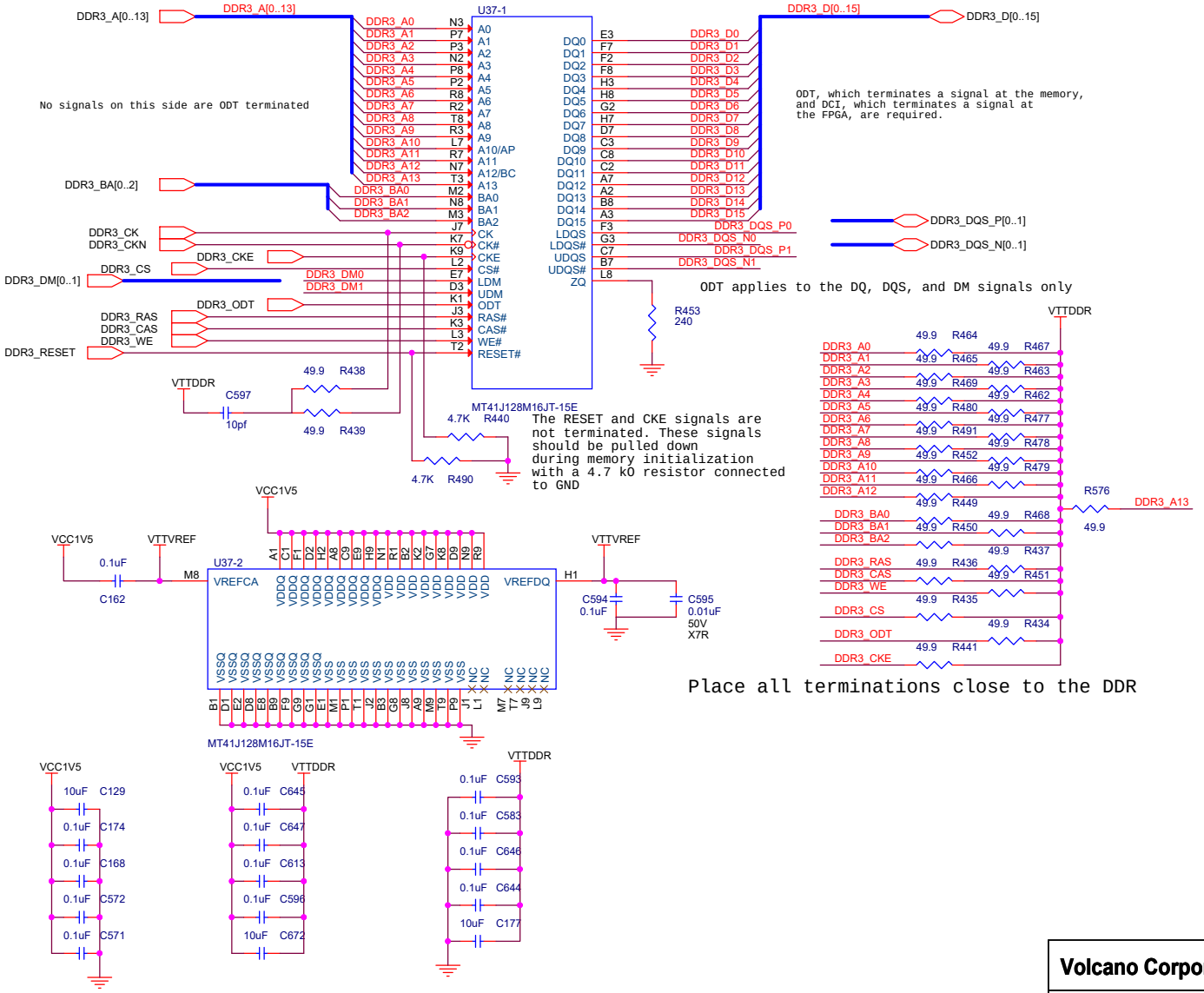
Volcano Corporation	2870 Kilgore Road, Rancho Cordova, CA 95670
Designer = Rc Ziegenbein	
Title: 3000-031-51102 ACE-e DDR3	
Size: B	Document Number: D000478762 — Rev: E
Date: Friday, December 30, 2022	Sheet 3 of 40



Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = Rc Ziegenbein		
Title 3000-031-51102 ACE-e DDR3		
Size B	Document Number D000478762	Rev E
Date: Friday, December 30, 2022	Sheet 4 of 40	



Volcano Corporation	2870 Kilgore Road, Rancho Cordova, CA 95670
Designer = Rc Ziegenbein	
Title: 3000-031-51102 ACE-e DDR3	
Size: B	Document Number: D000478762
Rev: E	
Date: Friday, December 30, 2022	Sheet 5 of 40



Volcano Corporation	2870 Kilgore Road, Rancho Cordova, CA 95670
Designer = Rc Ziegenbein	
Title: 3000-031-51102 ACE-e DDR3	
Size: B	Document Number: D000478762 — Rev: E
Date: Friday, December 30, 2022	Sheet 6 of 40

4 ddr3 chips 128M X 16 = 128M X64

DDR3E_DQS_N[0..7]
DDR3E_DQS_P[0..7]
DDR3E_DM[0..7]
DDR3E_D[0..63]

ASSY1
SDRAM SO-DIMM MT4JSF12864HZ-1G4D1

J5A

DDR3E_D4	4	DQ4	DQ32	129	DDR3E_D32
DDR3E_D0	5	DQ0	DQ36	130	DDR3E_D36
DDR3E_D5	6	DQ5	DQ33	131	DDR3E_D33
DDR3E_D1	7	DQ1	DQ37	132	DDR3E_D37
DDR3E_DQS_N0	10	DQS0_N	DSQ4_N	135	DDR3E_DQS_N4
DDR3E_DM0	11	DM0	DM4	136	DDR3E_DM4
DDR3E_DQS_P0	12	DQS0_P	DSQ4_P	137	DDR3E_DQS_P4
DDR3E_D2	15	DQ2	DQ38	140	DDR3E_D38
DDR3E_D6	16	DQ6	DQ34	141	DDR3E_D34
DDR3E_D3	17	DQ3	DQ39	142	DDR3E_D39
DDR3E_D7	18	DQ7	DQ35	143	DDR3E_D35
DDR3E_D8	21	DQ8	DQ44	146	DDR3E_D44
DDR3E_D12	22	DQ12	DQ40	147	DDR3E_D40
DDR3E_D9	23	DQ9	DQ45	148	DDR3E_D45
DDR3E_D13	24	DQ13	DQ41	149	DDR3E_D41
DDR3E_DQS_N1	27	DQS1_N	DQS5_N	152	DDR3E_DQS_N5
DDR3E_DM1	28	DM1	DM5	153	DDR3E_DM5
DDR3E_DQS_P1	29	DQS1_P	DQS5_P	154	DDR3E_DQS_P5
DDR3E_D10	33	DQ10	DQ42	157	DDR3E_D42
DDR3E_D14	34	DQ14	DQ46	158	DDR3E_D46
DDR3E_D11	35	DQ11	DQ43	159	DDR3E_D43
DDR3E_D15	36	DQ15	DQ47	160	DDR3E_D47
DDR3E_D16	39	DQ16	DQ48	163	DDR3E_D48
DDR3E_D20	40	DQ20	DQ52	164	DDR3E_D52
DDR3E_D17	41	DQ17	DQ49	165	DDR3E_D49
DDR3E_D21	42	DQ21	DQ53	166	DDR3E_D53
DDR3E_DQS_N2	45	DQS2_N	DQS6_N	169	DDR3E_DQS_N6
DDR3E_DM2	46	DM2	DM6	170	DDR3E_DM6
DDR3E_DQS_P2	47	DQS2_P	DQS6_P	171	DDR3E_DQS_P6
DDR3E_D22	50	DQ22	DQ54	174	DDR3E_D54
DDR3E_D18	51	DQ18	DQ50	175	DDR3E_D50
DDR3E_D23	52	DQ23	DQ55	176	DDR3E_D55
DDR3E_D19	53	DQ19	DQ51	177	DDR3E_D51
DDR3E_D28	56	DQ28	DQ60	180	DDR3E_D60
DDR3E_D24	57	DQ24	DQ56	181	DDR3E_D56
DDR3E_D29	58	DQ29	DQ61	182	DDR3E_D61
DDR3E_D25	59	DQ25	DQ57	183	DDR3E_D57
DDR3E_DQS_N3	62	DQS3_N	DQS7_N	186	DDR3E_DQS_N7
DDR3E_DM3	63	DM3	DM7	187	DDR3E_DM7
DDR3E_DQS_P3	64	DQS3_P	DQS7_P	188	DDR3E_DQS_P7
DDR3E_D26	67	DQ26	DQ58	191	DDR3E_D58
DDR3E_D30	68	DQ30	DQ62	192	DDR3E_D62
DDR3E_D27	69	DQ27	DQ59	193	DDR3E_D59
DDR3E_D31	70	DQ31	DQ63	194	DDR3E_D63

CONN SO-DIMM AS0A626-U2SN-7F

VTTDDR
VCC1V5
VTTVREF
Place Near
Pin 1 of DIMM

CN11 CN5 CN9: 0.1uF array X5R 10V
C362 C404 C375: 0.1uF 50 X7R
C370: 10uF 10V X7R
C423: 0.01uF 50V X7R
C341: 330uF 10V TANT

CN10 CN6 CN4: 0.1uF array X5R 10V
C363 C405 C415: 0.1uF 50 X7R
C403: 10uF 10V X7R
CN3 CN7 CN8: 0.1uF array X5R 10V
C395 C396 C385: 0.1uF 50 X7R
C353: 10uF 10V X7R

DDR3E_A[0..15]
DDR3E_BA[0..2]
DDR3E_RAS_B
DDR3E_CAS_B
DDR3E_WE_B
DDR3E_CLK_N[0..1]
DDR3E_CLK_P[0..1]
DDR3E_CKE[0..1]
DDR3E_ODT[0..1]
DDR3E_S0_B
DDR3E_S1_B
VTTDDR
VCC3V3
VTTVREF
.75 V ref voltage
.75 V termination voltage
R248 4.7K
R230 4.7K
DDR3E_CKE0
DDR3E_CKE1
DDR3E_RESET_B
R207 4.7K
R242 4.7K
R241 4.7K
R243 4.7K
R249 4.7K
R250 4.7K

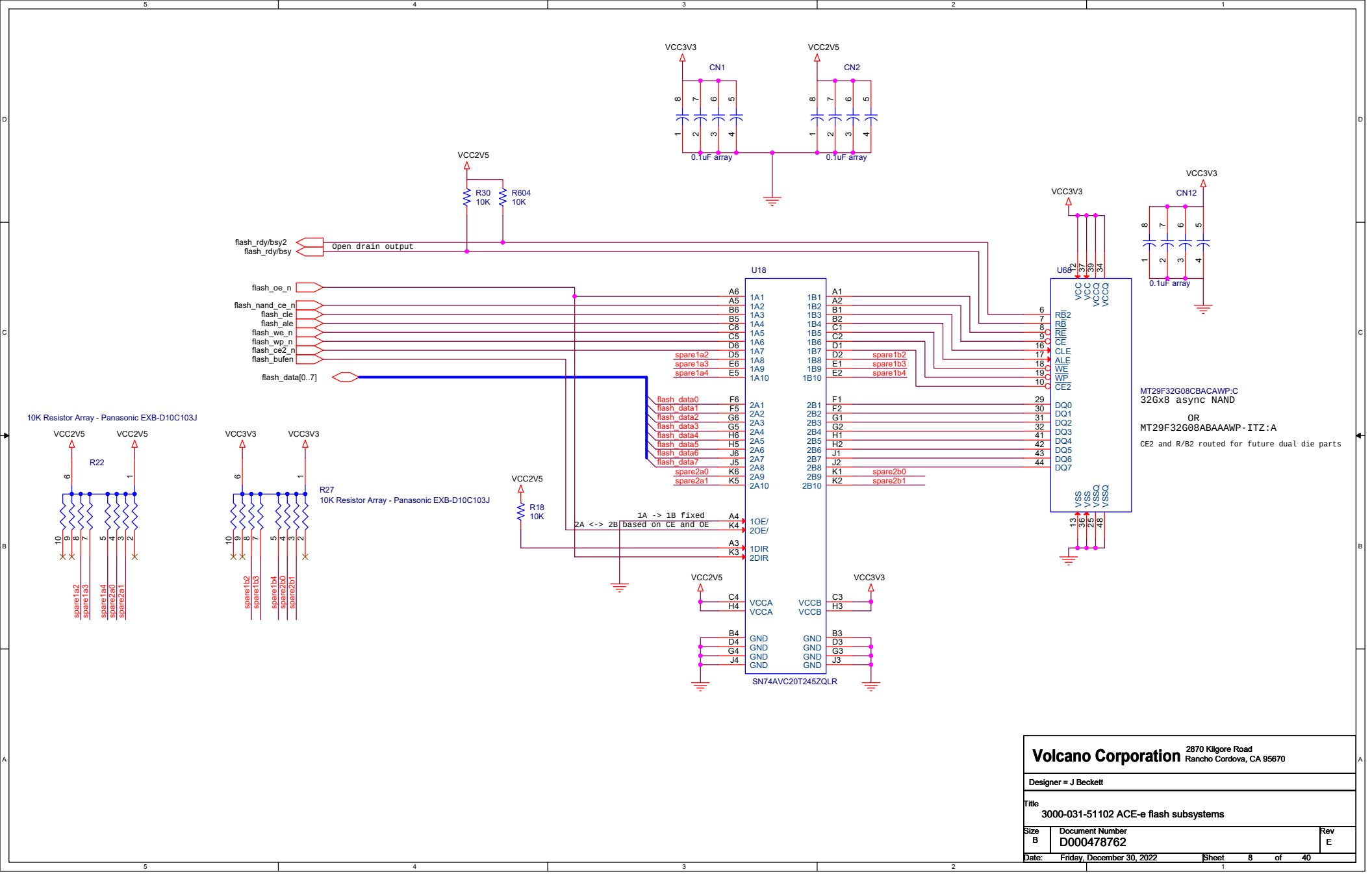
J5B

DDR3E_A0	98	A0	VDD1	75	VCC1V5
DDR3E_A1	97	A1	VDD2	76	
DDR3E_A2	96	A2	VDD3	81	
DDR3E_A3	95	A3	VDD4	82	
DDR3E_A4	92	A4	VDD5	87	
DDR3E_A5	91	A5	VDD6	88	
DDR3E_A6	90	A6	VDD7	93	
DDR3E_A7	86	A7	VDD8	94	
DDR3E_A8	89	A8	VDD9	99	
DDR3E_A9	85	A9	VDD10	100	
DDR3E_A10	107	A10/AP	VDD11	105	
DDR3E_A11	84	A11	VDD12	106	
DDR3E_A12	83	A12_BC_N	VDD13	111	
DDR3E_A13	119	A13	VDD14	112	
DDR3E_A14	80	A14	VDD15	117	
DDR3E_A15	78	A15	VDD16	118	
			VDD17	123	
			VDD18	124	
DDR3E_BA0	109	BA0	VSS1	2	
DDR3E_BA1	108	BA1	VSS2	3	
DDR3E_BA2	79	BA2	VSS3	8	
			VSS4	9	
			VSS5	13	
			VSS6	14	
			VSS7	19	
DDR3E_CLK_P0	101	CLK0_P	VSS8	20	
DDR3E_CLK_N0	103	CLK0_N	VSS9	25	
			VSS10	26	
DDR3E_CLK_P1	102	CLK1_P	VSS11	31	
DDR3E_CLK_N1	104	CLK1_N	VSS12	32	
			VSS13	37	
DDR3E_CKE0	73	CKE0	VSS14	38	
DDR3E_CKE1	74	CKE1	VSS15	43	
			VSS16	44	
DDR3E_ODT0	116	ODT0	VSS17	48	
DDR3E_ODT1	120	ODT1	VSS18	49	
			VSS19	54	
DDR3E_S0_B	114	S0_B	VSS20	55	
DDR3E_S1_B	121	S1_B	VSS21	60	
			VSS22	61	
	1	VREFDQ	VSS23	65	
	126	VREFCA	VSS24	66	
	199	VddSPD3.3	VSS25	71	
			VSS26	72	
	203	VTT1	VSS27	127	
	204	VTT2	VSS28	128	
			VSS29	133	
			VSS30	134	
			VSS31	138	
			VSS32	139	
			VSS33	144	
			VSS34	145	
			VSS35	150	
			VSS36	151	
			VSS37	155	
			VSS38	156	
			VSS39	161	
			VSS40	162	
			VSS41	167	
			VSS42	168	
			VSS43	172	
			VSS44	173	
			VSS45	178	
			VSS46	179	
			VSS47	184	
			VSS48	185	
			VSS49	189	
			VSS50	190	
			VSS51	195	
			VSS52	196	
DDR3E_RAS_B	110	RAS_B			
DDR3E_CAS_B	115	CAS_B			
DDR3E_WE_B	113	WE_B			
	125	Test /NC			
	77	NC1			
	122	NC2			
	198	Event_B			
	200	SDA			
	202	SCL			
	197	SA0			
	201	SA1			

CONN SO-DIMM AS0A626-U2SN-7F

DDR3E_TEMP_EVENT

Volcano Corp	2870 Kilgore	Rancho Cordova, CA 95670	Rc Ziegenbein
Title: Connector for SO-DIMM DDR3 RAM 128M X 64			
Size: B	Document Number: D000478762		Rev: E
Date: Friday, December 30, 2022	Sheet 7 of 40		



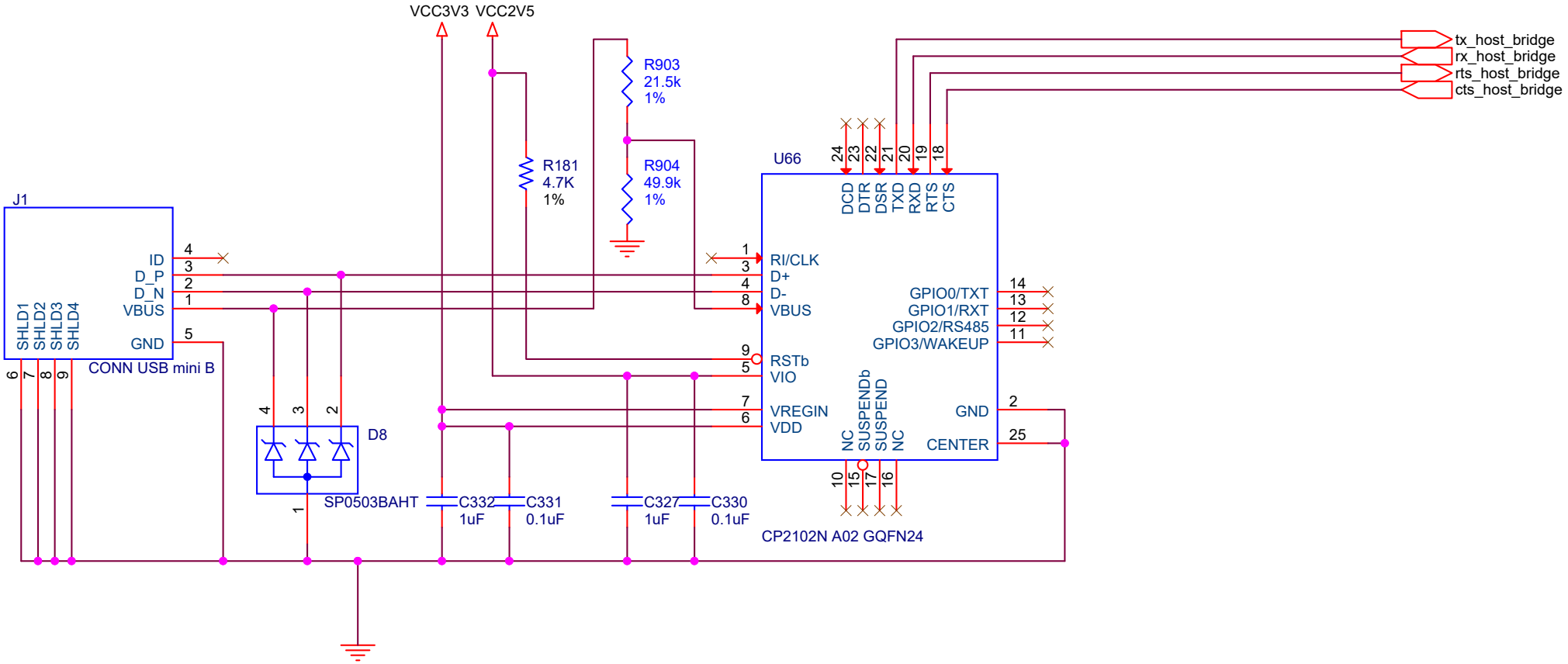
Volcano Corporation 2870 Kilgore Road, Rancho Cordova, CA 95670

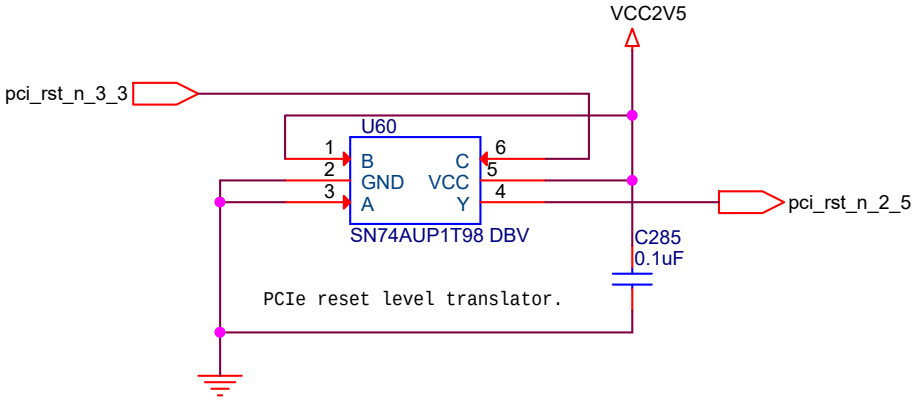
Designer = J Beckett

Title: 3000-031-51102 ACE-e flash subsystems

Size	Document Number	Rev
B	D000478762	E

Date: Friday, December 30, 2022 Sheet 8 of 40





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Rancho Cordova, CA 95670

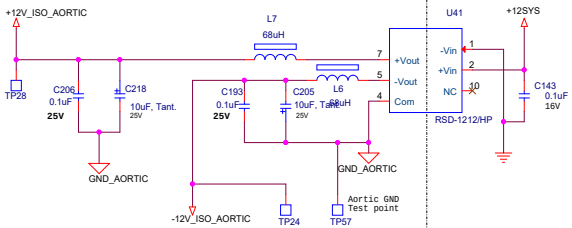
Designer = J Beckett

Title
3000-031-51102 ACE-e USB-UART bridge, FPGA/CPU reset managers

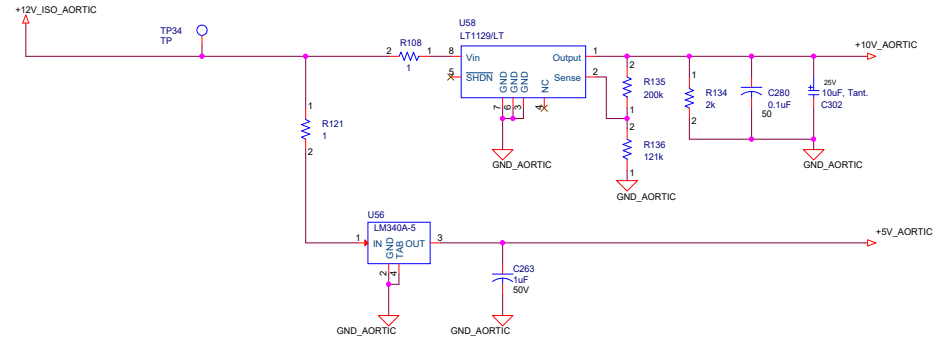
Size	Document Number	Rev
A	D000478762	E

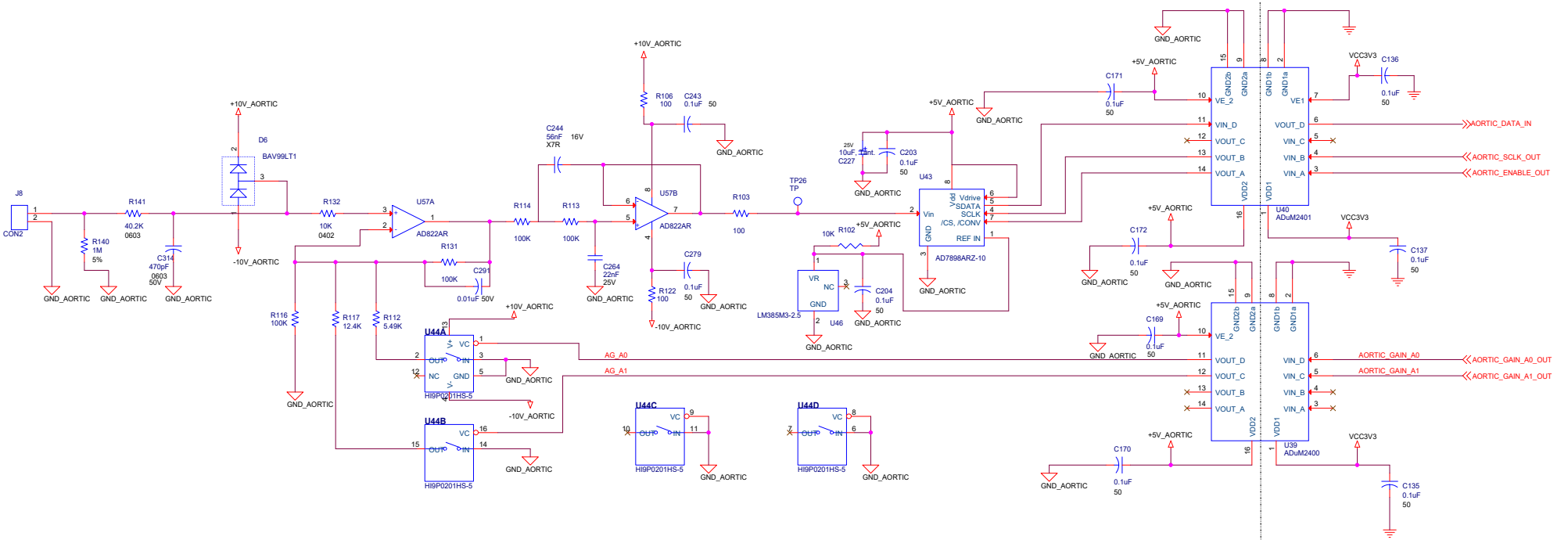
Date: Friday, December 30, 2022 Sheet 9 of 40

*ISOLATED POWER SUPPLY GENERATION - AORTIC



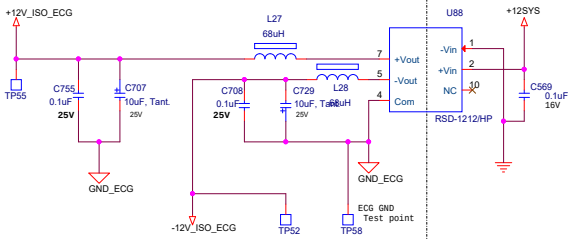
ISOLATED - AORTIC



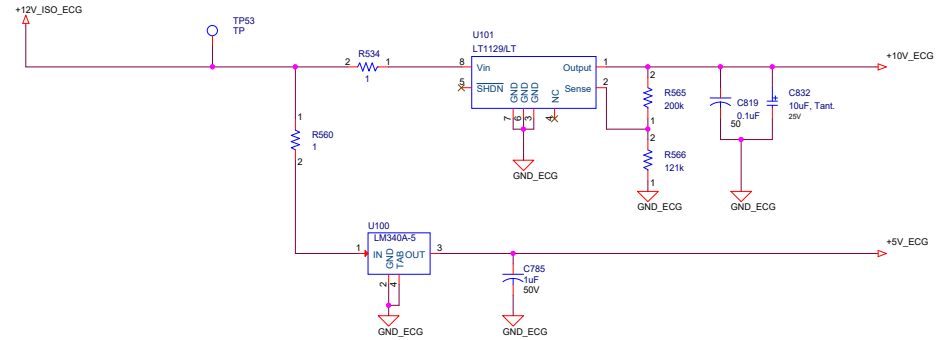


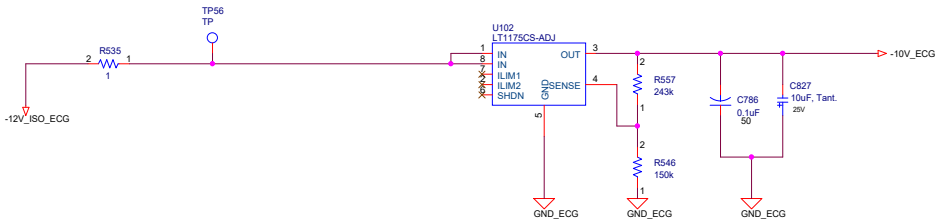
Title	3000-031-51102 ACE-e PCA - AORTIC INTERFACE/ISO POWER	
Size C	Document Number D0004787626	Rev E
Date: Friday, December 30, 2022	Sheet 10 of 40	

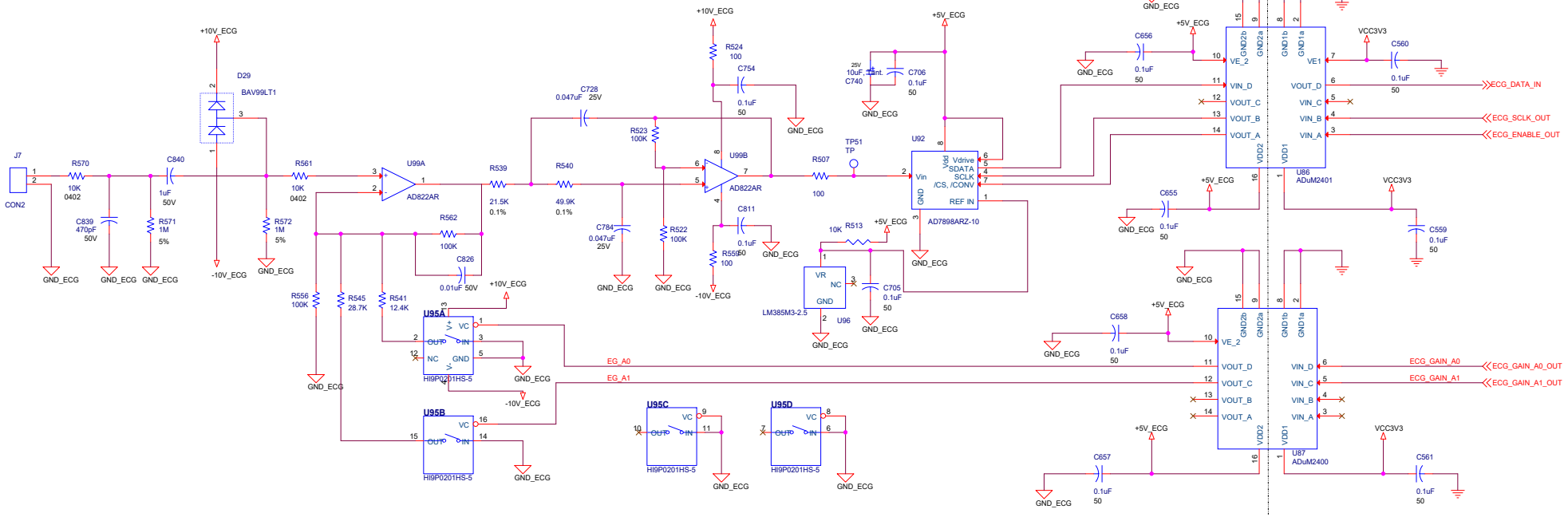
*ISOLATED POWER SUPPLY GENERATION - ECG



ISOLATED - ECG

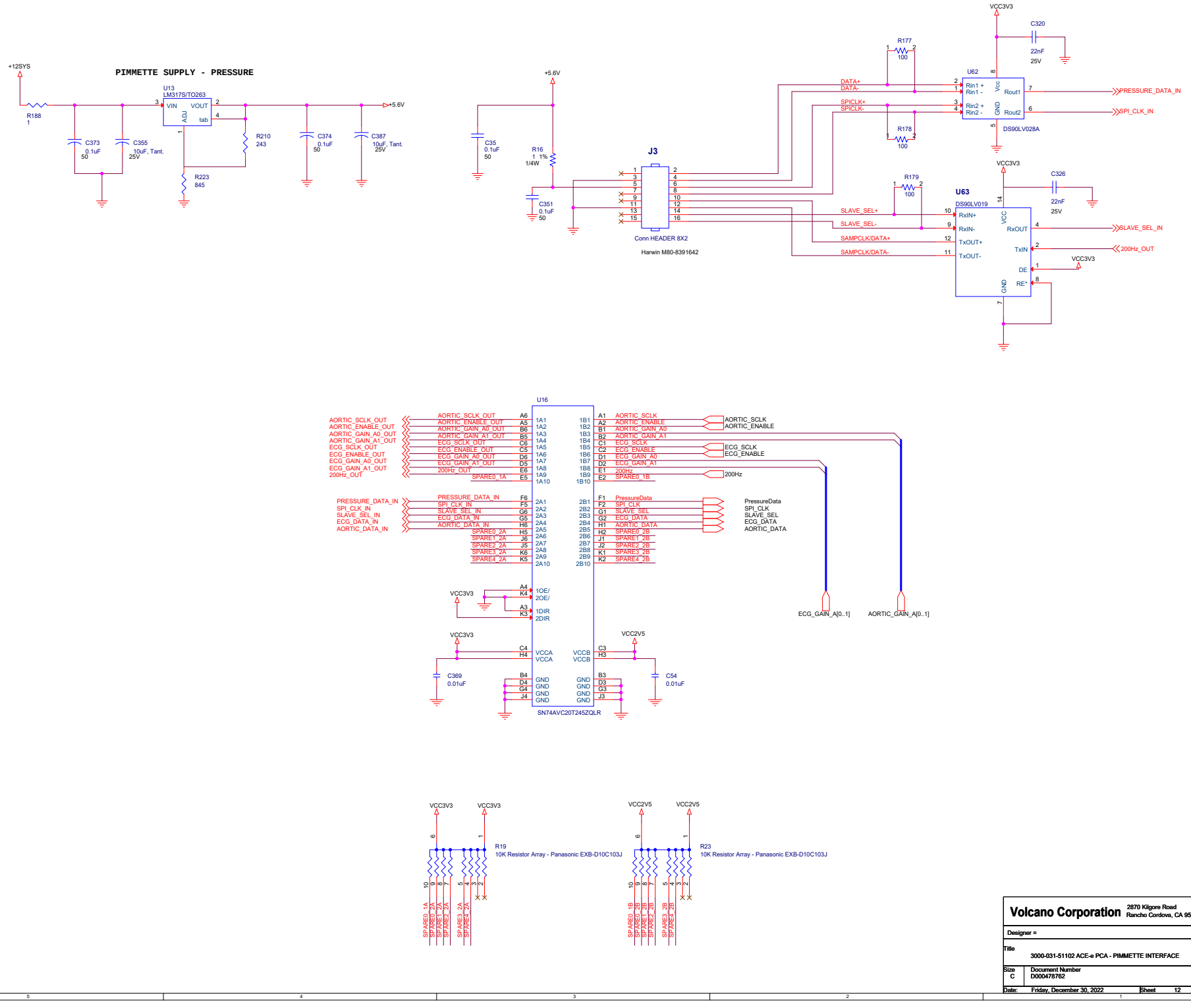


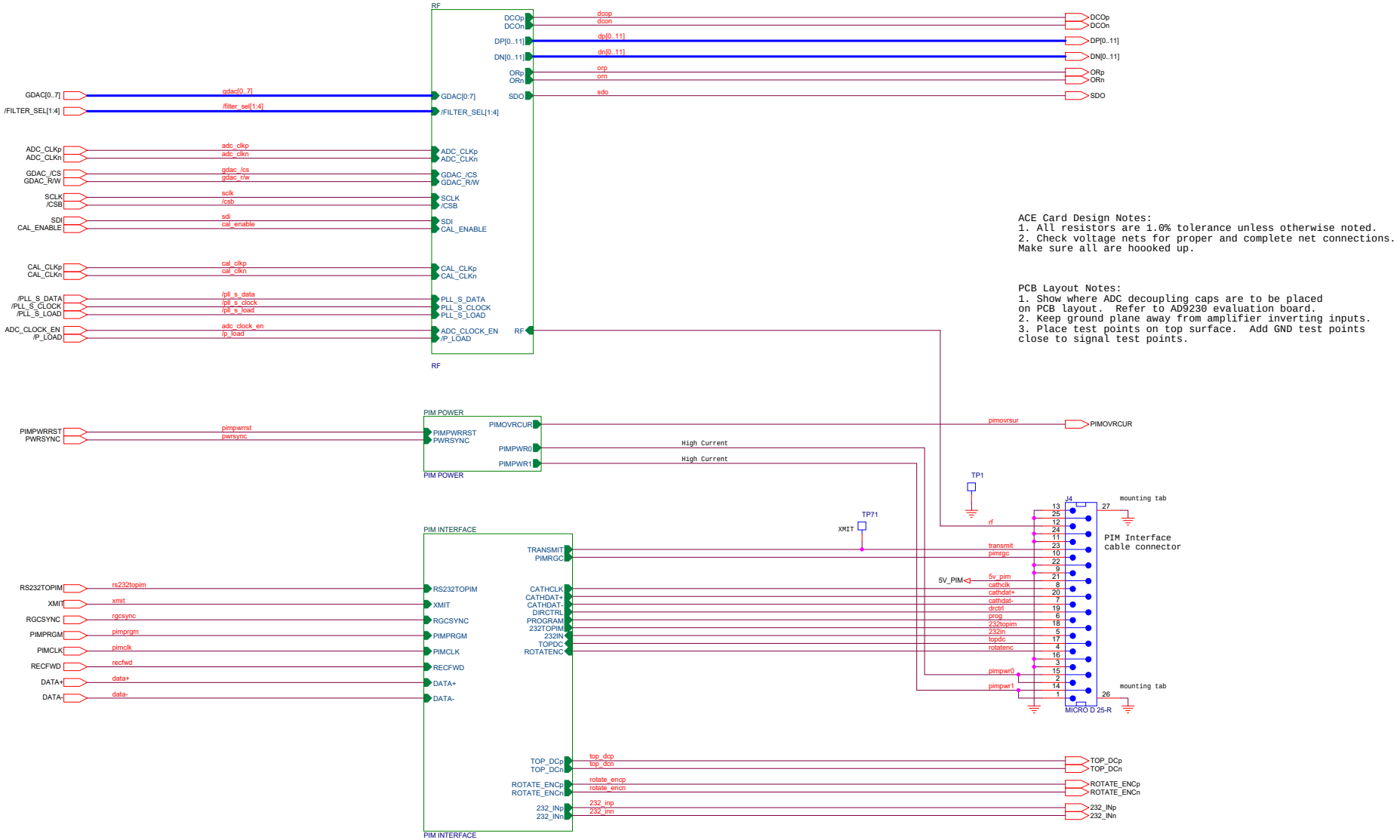




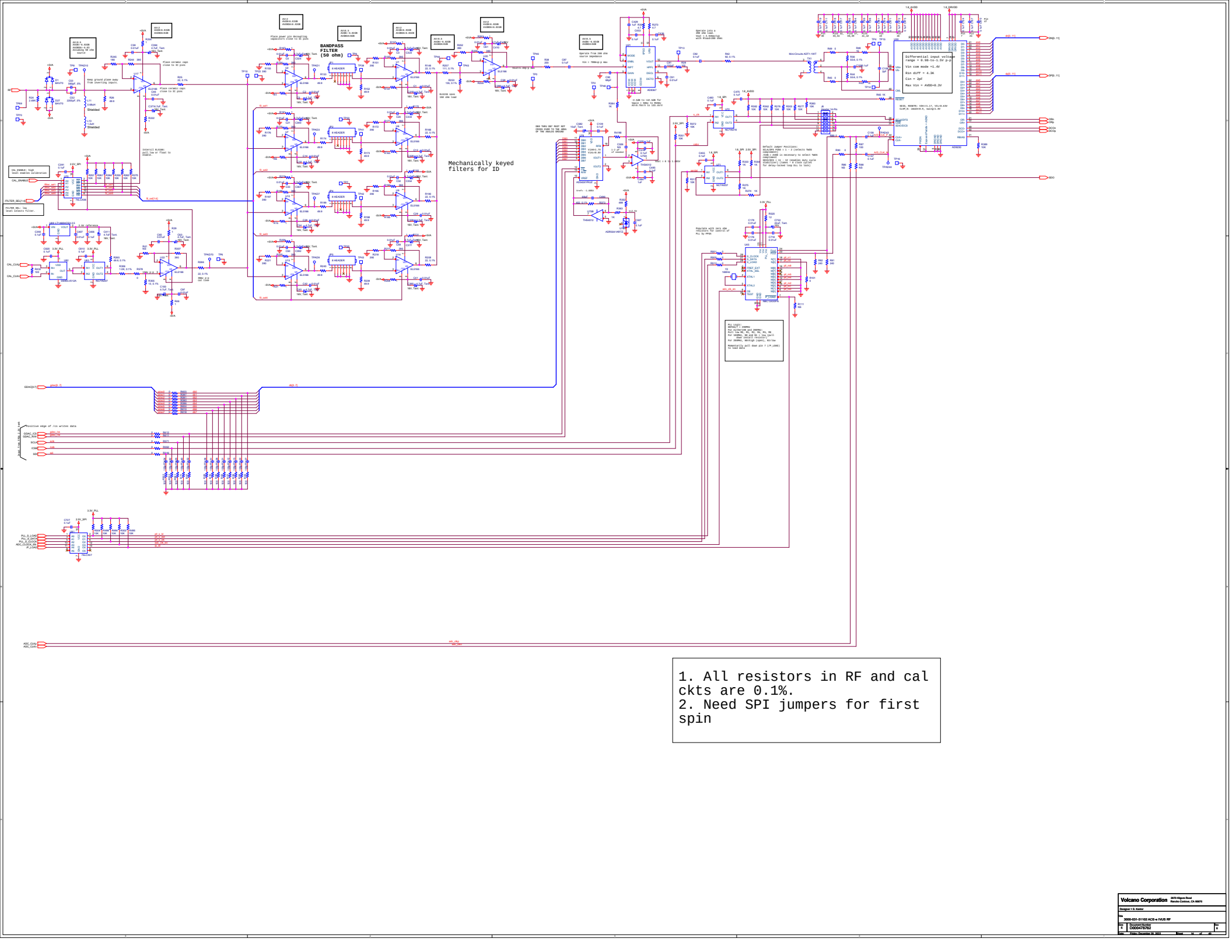
Volcano Corporation 2870 Kilgore Road Rancho Cordova, CA 95670

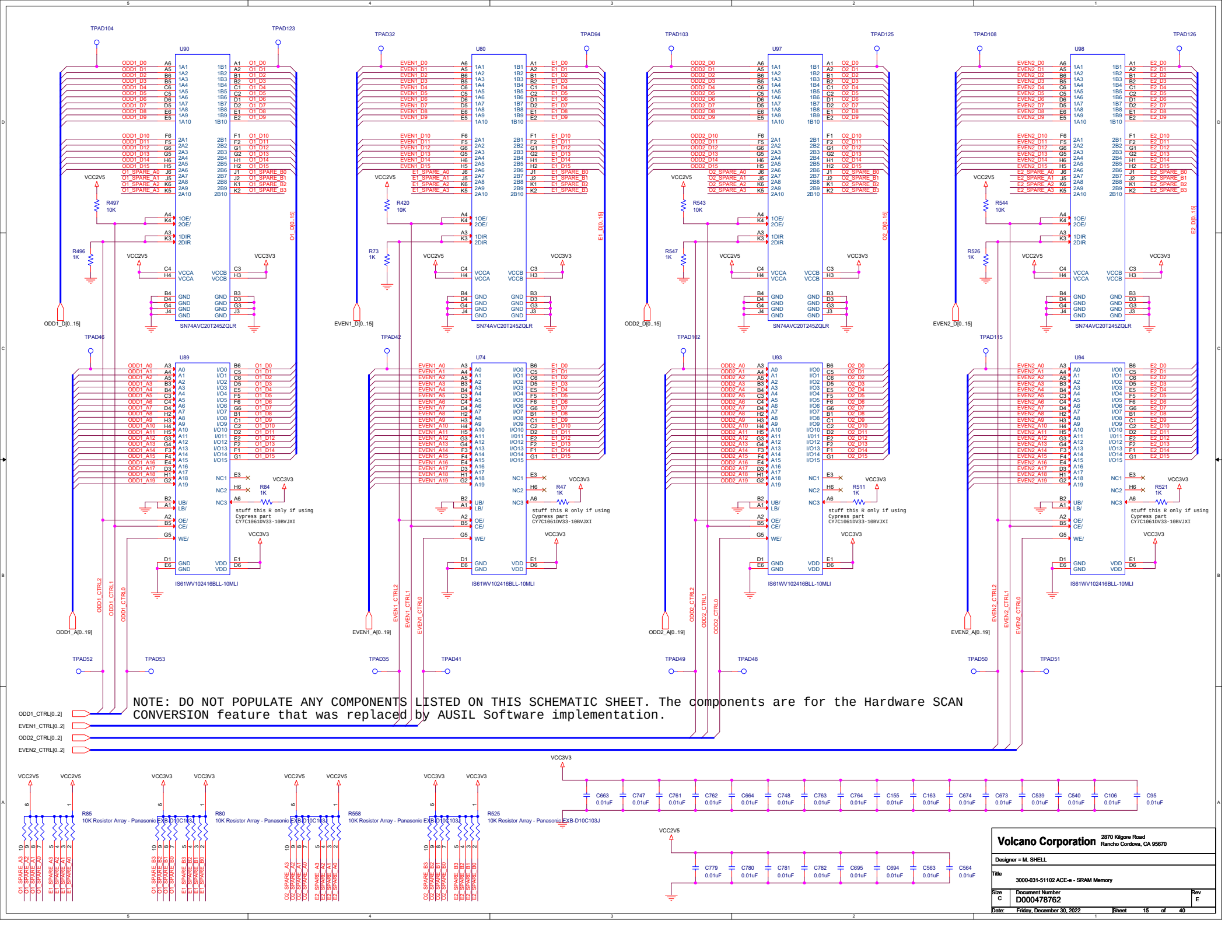
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Size C	Document Number D000478762 Rev E
Date: Friday, December 30, 2022	Sheet 11 of 40

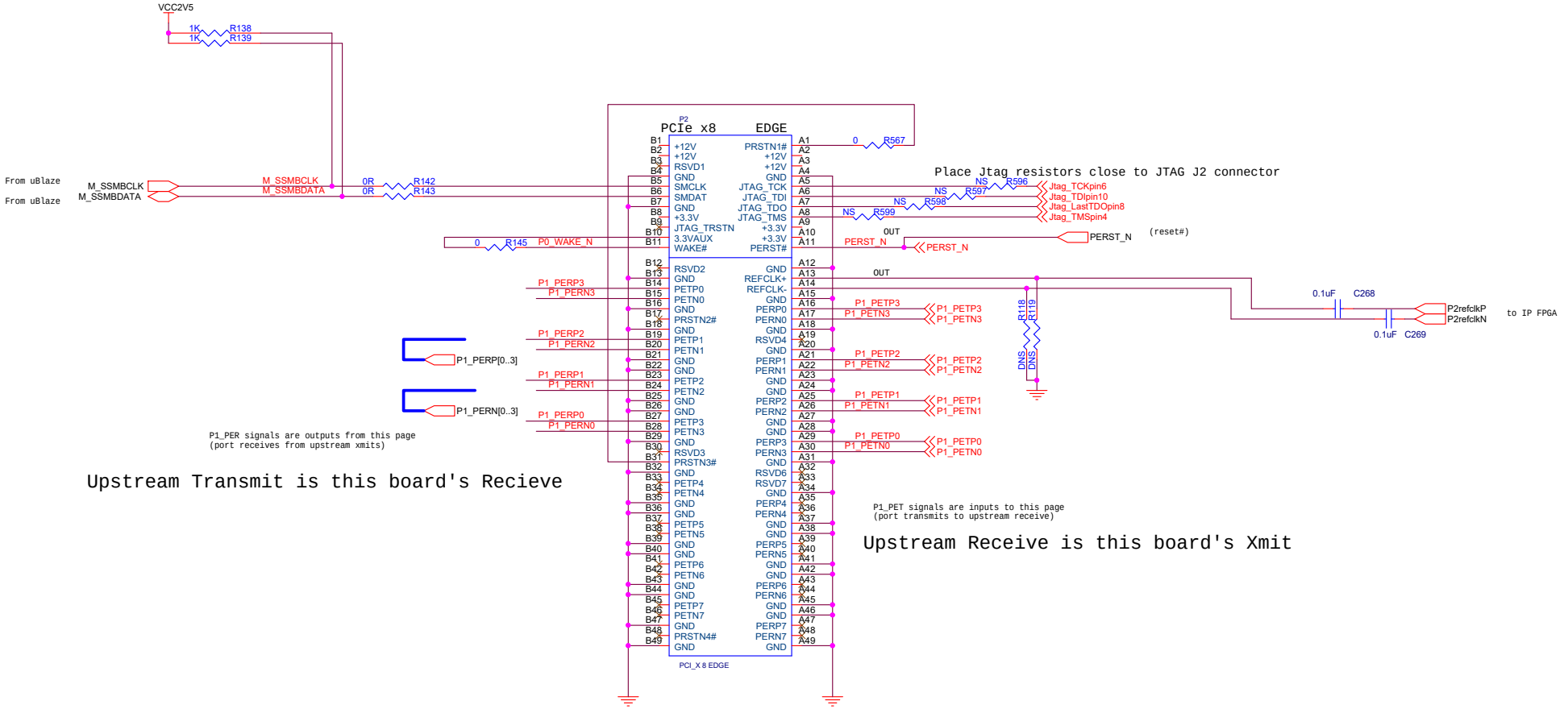

PIMMETTE SUPPLY - PRESSURE
+12SYS
R188
1
U13
LM317S/TO263
VIN
VOUT
ADJ
tab
C373
0.1uF
50
C355
10uF, Tant.
25V
R210
243
R223
845
C374
0.1uF
50
C387
10uF, Tant.
25V
+5.6V
C35
0.1uF
50
R16
1 1%
1/4W
C351
0.1uF
50
J3
Conn HEADER 8X2
Harwin M80-8391642
VCC3V3
C320
22nF
25V
R177
100
U62
DS90LV028A
Rin1 +
Rin1 -
Rin2 +
Rin2 -
Rout1
Rout2
Vcc
GND
DATA+
DATA-
SPICLK+
SPICLK-
PRESSURE_DATA_IN
SPI_CLK_IN
R178
100
R179
100
U63
DS90LV019
RxIN+
RxIN-
TxOUT+
TxOUT-
RxOUT
TxIN
DE
RE*
VCC
GND
SLAVE_SEL+
SLAVE_SEL-
SAMPCLK/DATA+
SAMPCLK/DATA-
C326
22nF
25V
SLAVE_SEL_IN
200Hz_OUT
U16
SN74AVC20T245ZQLR
AORTIC_SCLK_OUT
AORTIC_ENABLE_OUT
AORTIC_GAIN_A0_OUT
AORTIC_GAIN_A1_OUT
ECG_SCLK_OUT
ECG_ENABLE_OUT
ECG_GAIN_A0_OUT
ECG_GAIN_A1_OUT
200Hz_OUT
SPARE0_1A
PRESSURE_DATA_IN
SPI_CLK_IN
SLAVE_SEL_IN
ECG_DATA_IN
AORTIC_DATA_IN
SPARE0_2A
SPARE1_2A
SPARE2_2A
SPARE3_2A
SPARE4_2A
SPARE0_1B
SPARE0_2B
SPARE1_2B
SPARE2_2B
SPARE3_2B
SPARE4_2B
AORTIC_SCLK
AORTIC_ENABLE
AORTIC_GAIN_A0
AORTIC_GAIN_A1
ECG_SCLK
ECG_ENABLE
ECG_GAIN_A0
ECG_GAIN_A1
200Hz
PressureData
SPI_CLK
SLAVE_SEL
ECG_DATA
AORTIC_DATA
ECG_GAIN_A[0..1]
AORTIC_GAIN_A[0..1]
1OE/
2OE/
1DIR
2DIR
VCCA
VCCB
GND
C369
0.01uF
C54
0.01uF
VCC2V5
R19
10K Resistor Array - Panasonic EXB-D10C103J
R23
10K Resistor Array - Panasonic EXB-D10C103J
Volcano Corporation
2870 Kilgore Road
Rancho Cordova, CA 95670
Designer =
Title
3000-031-51102 ACE-e PCA - PIMMETTE INTERFACE
Size
C
Document Number
D000478762
Rev
E
Date:
Friday, December 30, 2022
Sheet
12
of
40



Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = S. Kantor		
Title: 3000-031-51102 ACE-e IVUS Analog Top		
Size: C	Document Number: D000478762	Rev: E
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BANDPASS FILTER (50 ohm)
Mechanically keyed filters for ID
1. All resistors in RF and cal ckts are 0.1%.
2. Need SPI jumpers for first spin
Volcano Corporation
3870 Kilgore Road
Rancho Cordova, CA 95670
Designer = S. Kiefer
Title
3000-031-01102 ACE-e IVUS RF
Size
A
Document Number
D00047876Z
Rev
E
Date: Friday, December 30, 2011
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TPAD104
TPAD123
U90
ODD1_D0
ODD1_D1
ODD1_D2
ODD1_D3
ODD1_D4
ODD1_D5
ODD1_D6
ODD1_D7
ODD1_D8
ODD1_D9
ODD1_D10
ODD1_D11
ODD1_D12
ODD1_D13
ODD1_D14
ODD1_D15
O1_SPARE_A0
O1_SPARE_A1
O1_SPARE_A2
O1_SPARE_A3
O1_D0
O1_D1
O1_D2
O1_D3
O1_D4
O1_D5
O1_D6
O1_D7
O1_D8
O1_D9
O1_D10
O1_D11
O1_D12
O1_D13
O1_D14
O1_D15
O1_SPARE_B0
O1_SPARE_B1
O1_SPARE_B2
O1_SPARE_B3
VCC2V5
VCC3V3
R497
10K
R496
1K
1OE/
2OE/
1DIR
2DIR
VCCA
VCCB
GND
SN74AVC20T245ZQLR
ODD1_D[0..15]
O1_D[0..15]
TPAD46
U89
ODD1_A0
ODD1_A1
ODD1_A2
ODD1_A3
ODD1_A4
ODD1_A5
ODD1_A6
ODD1_A7
ODD1_A8
ODD1_A9
ODD1_A10
ODD1_A11
ODD1_A12
ODD1_A13
ODD1_A14
ODD1_A15
ODD1_A16
ODD1_A17
ODD1_A18
ODD1_A19
R84
1K
stuff this R only if using Cypress part CY7C1061DV33-10BVJXI
UB/
LB/
OE/
CE/
WE/
NC1
NC2
NC3
VDD
IS61WV102416BLL-10MLI
ODD1_A[0..19]
ODD1_CTRL2
ODD1_CTRL1
ODD1_CTRL0
TPAD52
TPAD53
TPAD32
TPAD94
U80
EVEN1_D0
EVEN1_D1
EVEN1_D2
EVEN1_D3
EVEN1_D4
EVEN1_D5
EVEN1_D6
EVEN1_D7
EVEN1_D8
EVEN1_D9
EVEN1_D10
EVEN1_D11
EVEN1_D12
EVEN1_D13
EVEN1_D14
EVEN1_D15
E1_SPARE_A0
E1_SPARE_A1
E1_SPARE_A2
E1_SPARE_A3
E1_D0
E1_D1
E1_D2
E1_D3
E1_D4
E1_D5
E1_D6
E1_D7
E1_D8
E1_D9
E1_D10
E1_D11
E1_D12
E1_D13
E1_D14
E1_D15
E1_SPARE_B0
E1_SPARE_B1
E1_SPARE_B2
E1_SPARE_B3
R420
10K
R73
1K
EVEN1_D[0..15]
E1_D[0..15]
TPAD42
U74
EVEN1_A0
EVEN1_A1
EVEN1_A2
EVEN1_A3
EVEN1_A4
EVEN1_A5
EVEN1_A6
EVEN1_A7
EVEN1_A8
EVEN1_A9
EVEN1_A10
EVEN1_A11
EVEN1_A12
EVEN1_A13
EVEN1_A14
EVEN1_A15
EVEN1_A16
EVEN1_A17
EVEN1_A18
EVEN1_A19
R47
1K
stuff this R only if using Cypress part CY7C1061DV33-10BVJXI
IS61WV102416BLL-10MLI
EVEN1_A[0..19]
EVEN1_CTRL2
EVEN1_CTRL1
EVEN1_CTRL0
TPAD35
TPAD41
TPAD103
TPAD125
U97
ODD2_D0
ODD2_D1
ODD2_D2
ODD2_D3
ODD2_D4
ODD2_D5
ODD2_D6
ODD2_D7
ODD2_D8
ODD2_D9
ODD2_D10
ODD2_D11
ODD2_D12
ODD2_D13
ODD2_D14
ODD2_D15
O2_SPARE_A0
O2_SPARE_A1
O2_SPARE_A2
O2_SPARE_A3
O2_D0
O2_D1
O2_D2
O2_D3
O2_D4
O2_D5
O2_D6
O2_D7
O2_D8
O2_D9
O2_D10
O2_D11
O2_D12
O2_D13
O2_D14
O2_D15
O2_SPARE_B0
O2_SPARE_B1
O2_SPARE_B2
O2_SPARE_B3
R543
10K
R547
1K
ODD2_D[0..15]
O2_D[0..15]
TPAD102
U93
ODD2_A0
ODD2_A1
ODD2_A2
ODD2_A3
ODD2_A4
ODD2_A5
ODD2_A6
ODD2_A7
ODD2_A8
ODD2_A9
ODD2_A10
ODD2_A11
ODD2_A12
ODD2_A13
ODD2_A14
ODD2_A15
ODD2_A16
ODD2_A17
ODD2_A18
ODD2_A19
R511
1K
stuff this R only if using Cypress part CY7C1061DV33-10BVJXI
IS61WV102416BLL-10MLI
ODD2_A[0..19]
ODD2_CTRL2
ODD2_CTRL1
ODD2_CTRL0
TPAD49
TPAD48
TPAD108
TPAD126
U98
EVEN2_D0
EVEN2_D1
EVEN2_D2
EVEN2_D3
EVEN2_D4
EVEN2_D5
EVEN2_D6
EVEN2_D7
EVEN2_D8
EVEN2_D9
EVEN2_D10
EVEN2_D11
EVEN2_D12
EVEN2_D13
EVEN2_D14
EVEN2_D15
E2_SPARE_A0
E2_SPARE_A1
E2_SPARE_A2
E2_SPARE_A3
E2_D0
E2_D1
E2_D2
E2_D3
E2_D4
E2_D5
E2_D6
E2_D7
E2_D8
E2_D9
E2_D10
E2_D11
E2_D12
E2_D13
E2_D14
E2_D15
E2_SPARE_B0
E2_SPARE_B1
E2_SPARE_B2
E2_SPARE_B3
R544
10K
R526
1K
EVEN2_D[0..15]
E2_D[0..15]
TPAD115
U94
EVEN2_A0
EVEN2_A1
EVEN2_A2
EVEN2_A3
EVEN2_A4
EVEN2_A5
EVEN2_A6
EVEN2_A7
EVEN2_A8
EVEN2_A9
EVEN2_A10
EVEN2_A11
EVEN2_A12
EVEN2_A13
EVEN2_A14
EVEN2_A15
EVEN2_A16
EVEN2_A17
EVEN2_A18
EVEN2_A19
R521
1K
stuff this R only if using Cypress part CY7C1061DV33-10BVJXI
IS61WV102416BLL-10MLI
EVEN2_A[0..19]
EVEN2_CTRL2
EVEN2_CTRL1
EVEN2_CTRL0
TPAD50
TPAD51
NOTE: DO NOT POPULATE ANY COMPONENTS LISTED ON THIS SCHEMATIC SHEET. The components are for the Hardware SCAN CONVERSION feature that was replaced by AUSIL Software implementation.
ODD1_CTRL[0..2]
EVEN1_CTRL[0..2]
ODD2_CTRL[0..2]
EVEN2_CTRL[0..2]
R85
10K Resistor Array - Panasonic EXB-D10C103J
R80
10K Resistor Array - Panasonic EXB-D10C103J
R558
10K Resistor Array - Panasonic EXB-D10C103J
R525
10K Resistor Array - Panasonic EXB-D10C103J
C663 0.01uF
C747 0.01uF
C761 0.01uF
C762 0.01uF
C664 0.01uF
C748 0.01uF
C763 0.01uF
C764 0.01uF
C155 0.01uF
C163 0.01uF
C674 0.01uF
C673 0.01uF
C539 0.01uF
C540 0.01uF
C106 0.01uF
C95 0.01uF
C779 0.01uF
C780 0.01uF
C781 0.01uF
C782 0.01uF
C695 0.01uF
C694 0.01uF
C563 0.01uF
C564 0.01uF
Volcano Corporation
2870 Kilgore Road
Rancho Cordova, CA 95670
Designer = M. SHELL
Title
3000-031-51102 ACE-e - SRAM Memory
Size C
Document Number D000478762
Rev E
Date: Friday, December 30, 2022
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EDGE CONN. PCIe

Volcano Corporation 2870 Kilgore Road Rancho Cordova, CA 95670		
Designer = Rc Ziegenbein		
Title: 3000-031-51102 ACE-e PCIe interface		
Size: A3	Document Number: D000478762	Rev: E
Date: Friday, December 30, 2022	Sheet 16 of 40	

Port 1 = PCIe connector to upstream host

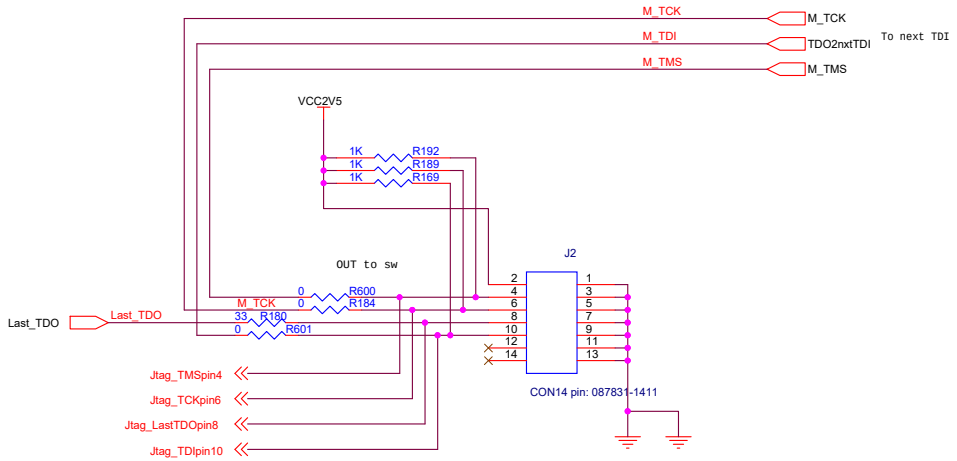
From IP FPGA

AC_P1_PETP[0..3]
AC_P1_PETN[0..3]

Outputs on this side

AC_P1_PETP0		0.1UF	C311	P1_PETP0	P1_PETP0	To PCIe edge connector
AC_P1_PETN0	0.1UF	C312		P1_PETN0	P1_PETN0	
AC_P1_PETP1		0.1UF	C309	P1_PETP1	P1_PETP1	
AC_P1_PETN1	0.1UF	C310		P1_PETN1	P1_PETN1	
AC_P1_PETP2		0.1UF	C307	P1_PETP2	P1_PETP2	
AC_P1_PETN2	0.1UF	C308		P1_PETN2	P1_PETN2	
AC_P1_PETP3		0.1UF	C305	P1_PETP3	P1_PETP3	
AC_P1_PETN3	0.1UF	C306		P1_PETN3	P1_PETN3	

NOTE:
PLACE ALL AC COUPLING CAPACITORS NEAR
PCIe EDGE CONNECTOR



Rev 008 mods 11-jun-2015 (jwb): Changed JTAG pullups to connect to 2.5V, not 3.3V. The Virtex-6 is not 3.3V compliant; this violated the VCCO + 0.2V max.

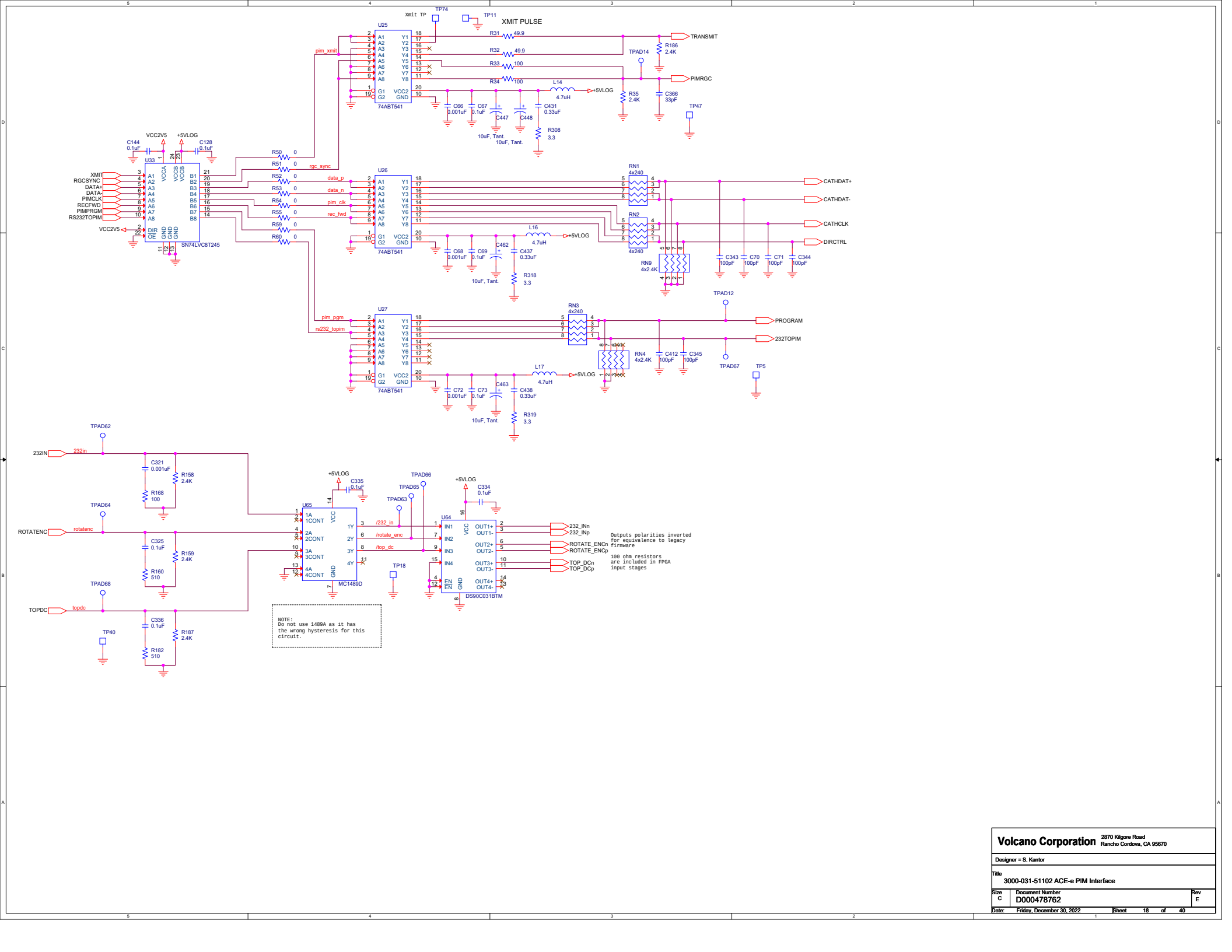
Volcano Corporation 2870 Kilgore Road, Rancho Cordova, CA 95670

Designer = Rc Ziegenbein

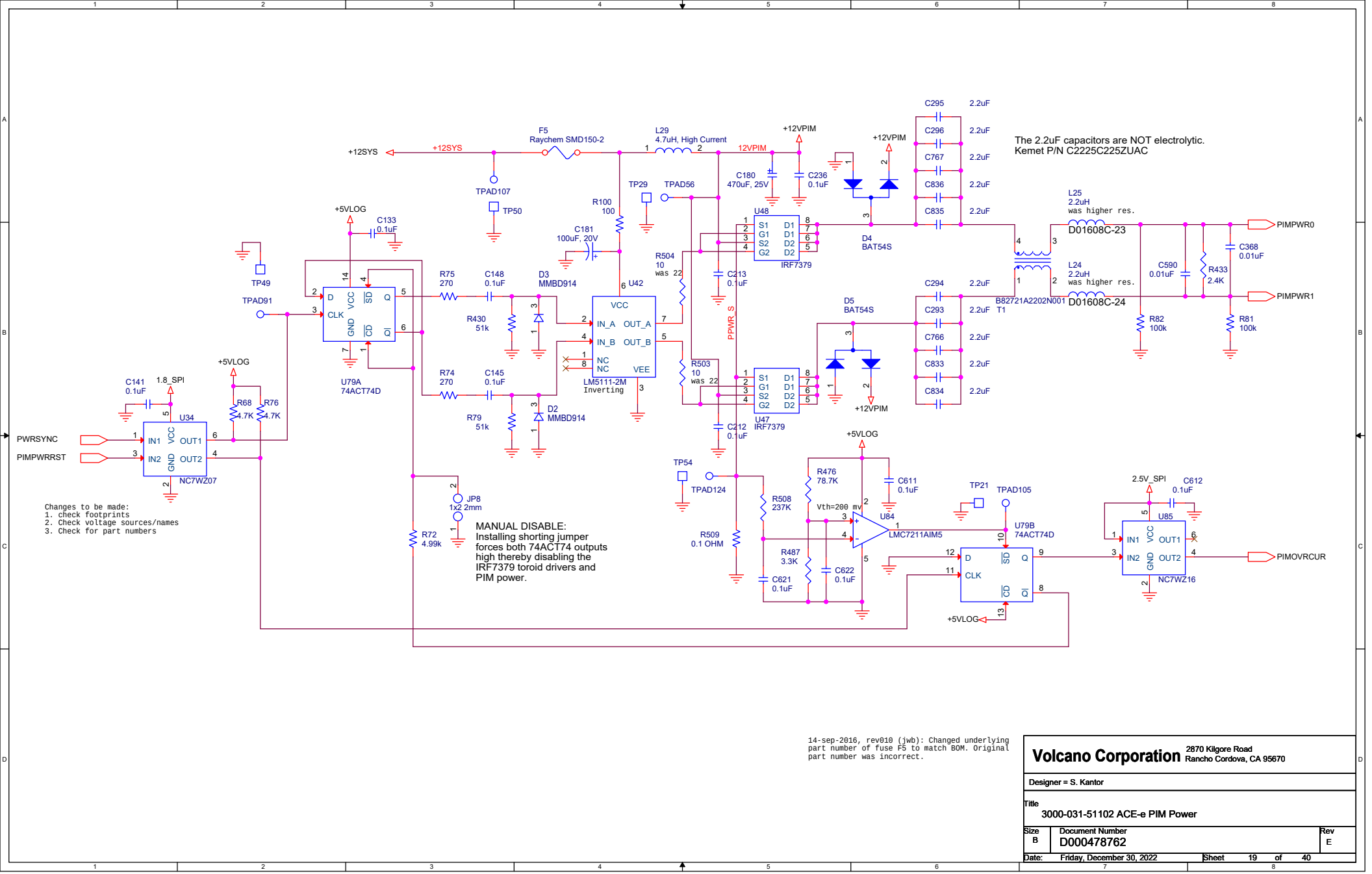
Title: 3000-031-51102 ACE-e PCIe switch

Size	Document Number	Rev
A3	D000478762	E

Date: Friday, December 30, 2022 Sheet 17 of 40

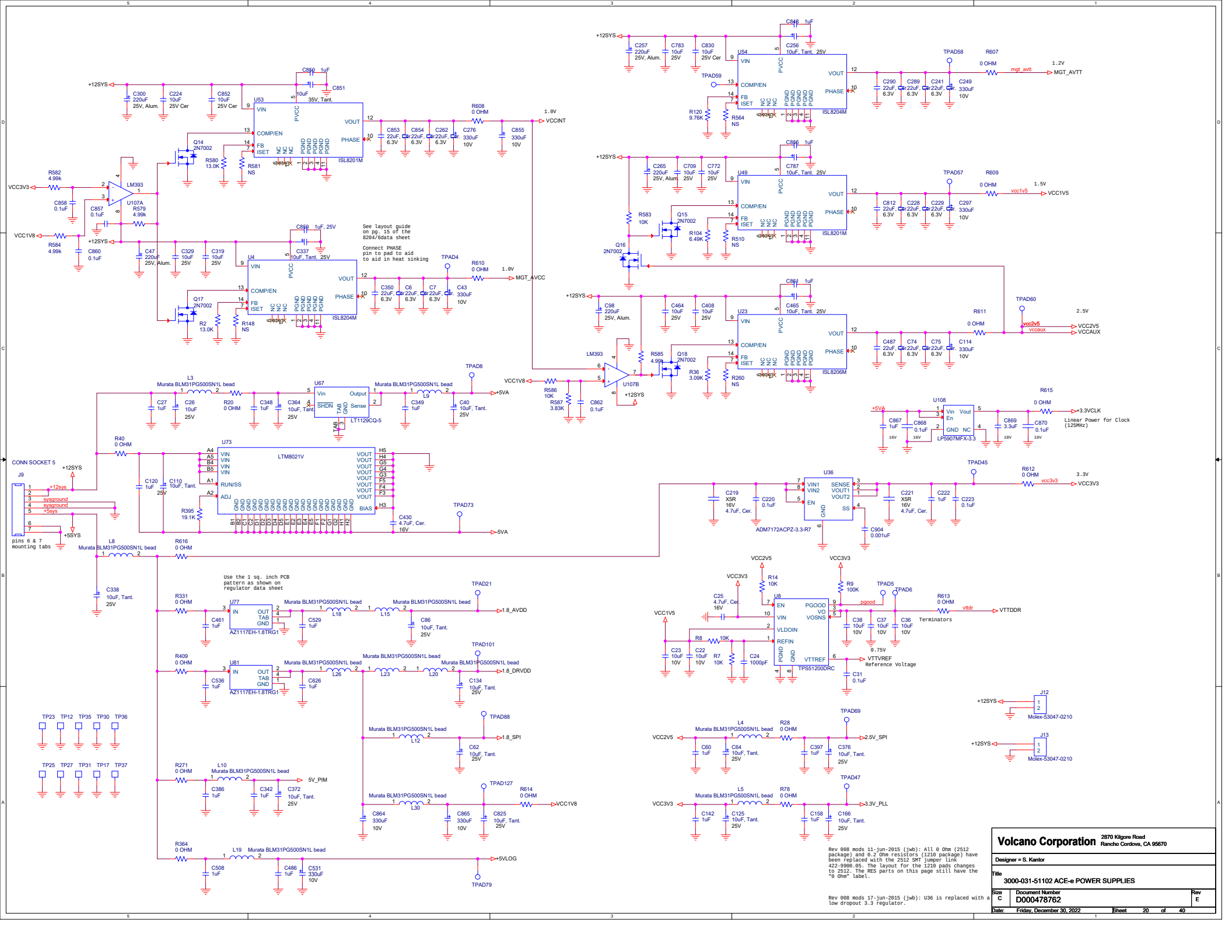


Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = S. Kantor		
Title: 3000-031-51102 ACE-e PIM Interface		
Size: C	Document Number: D000478762	Rev: E
Date: Friday, December 30, 2022	Sheet 18 of 40	



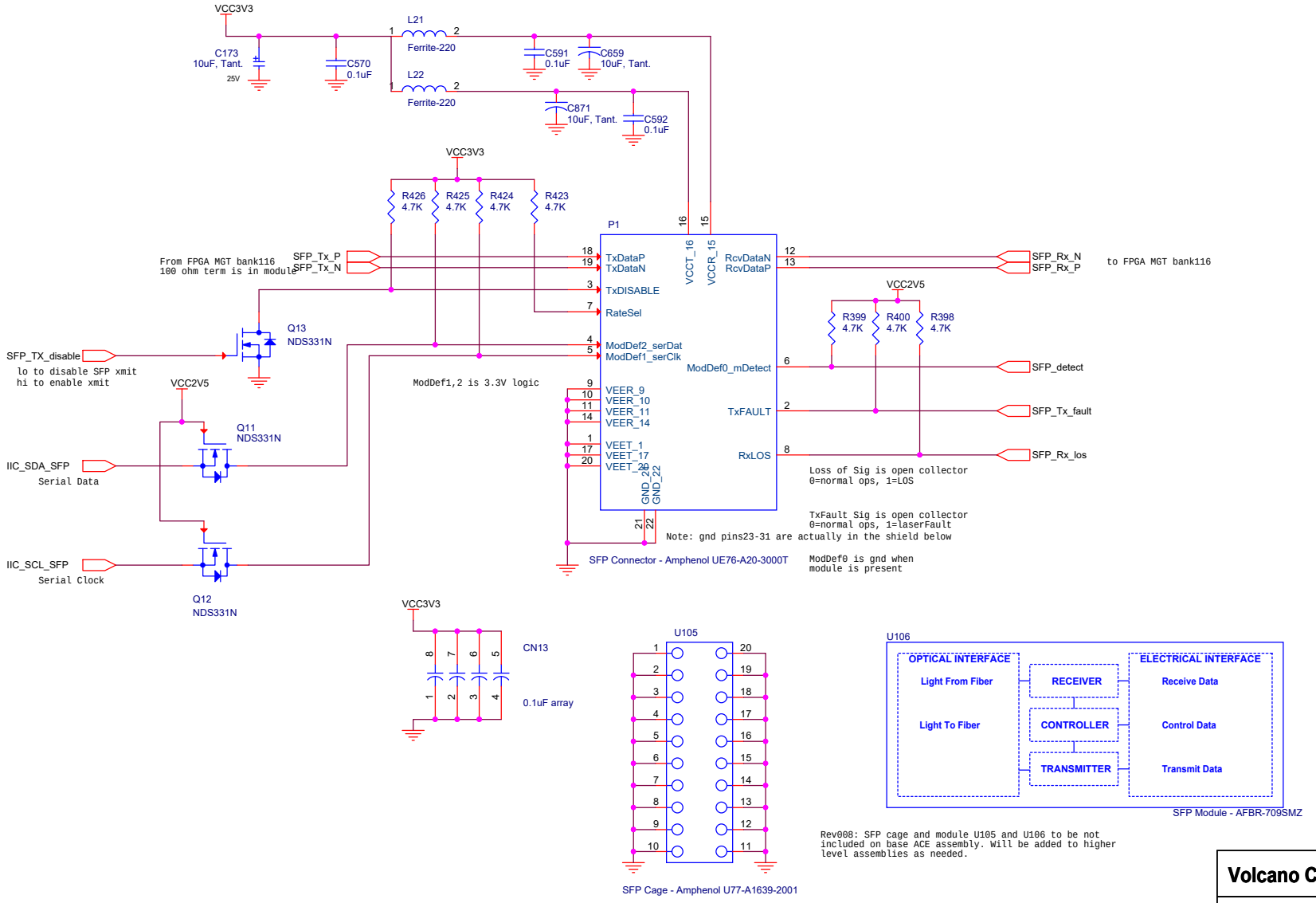
14-sep-2016, rev010 (jwb): Changed underlying part number of fuse F5 to match BOM. Original part number was incorrect.

Volcano Corporation 2870 Kilgore Road Rancho Cordova, CA 95670		
Designer = S. Kantor		
Title: 3000-031-51102 ACE-e PIM Power		
Size: B	Document Number: D000478762	Rev: E
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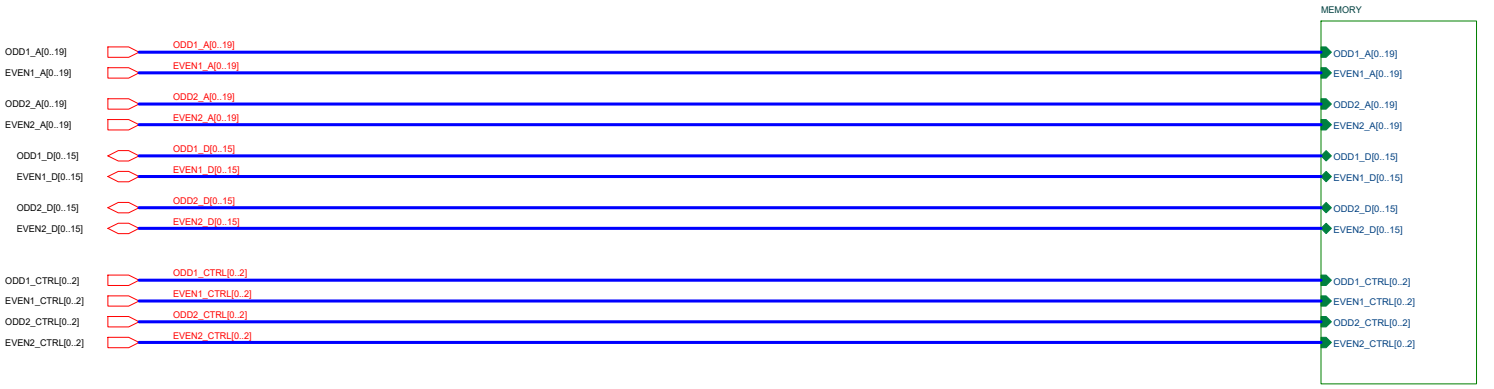


Rev 008 mods 11-jun-2015 (jwb): All 0 Ohm (2512 package) and 0.2 Ohm resistors (1210 package) have been replaced with the 2512 SMT jumper link 422-0900.05. The layout for the 1210 pads changes to 2512. The RES parts on this page still have the "0 Ohm" label.

Rev 008 mods 17-jun-2015 (jwb): U36 is replaced with a low dropout 3.3 regulator.



Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = Rc Ziegenbein		
Title 3000-031-51102 ACE-e SFP module		
Size B	Document Number D000478762	Rev E
Date: Friday, December 30, 2022	Sheet 21 of 40	



SHEET 3

TP38 TP19

Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = M. SHELL		
Title: 3000-031-51102 ACE-e Top SRAM Memory		
Size: C	Document Number: D000478762	Rev: E
Date: Friday, December 30, 2022	Sheet 22 of 40	

Master BPI up Mode=010
Slave SelectMAP Mode=110
bus width= 8, 16, 32
cclk= Input

hswapen=0: pullup I/O during config
rdwr_b=0: dir of D[15:0] =inputs
csi_b=0: selectMAP data bus enabled

If the thermal diode is not
being used, these pins
(DXP,DXN) should be
tied to ground

Bank 24

Caution, bank will be used for config

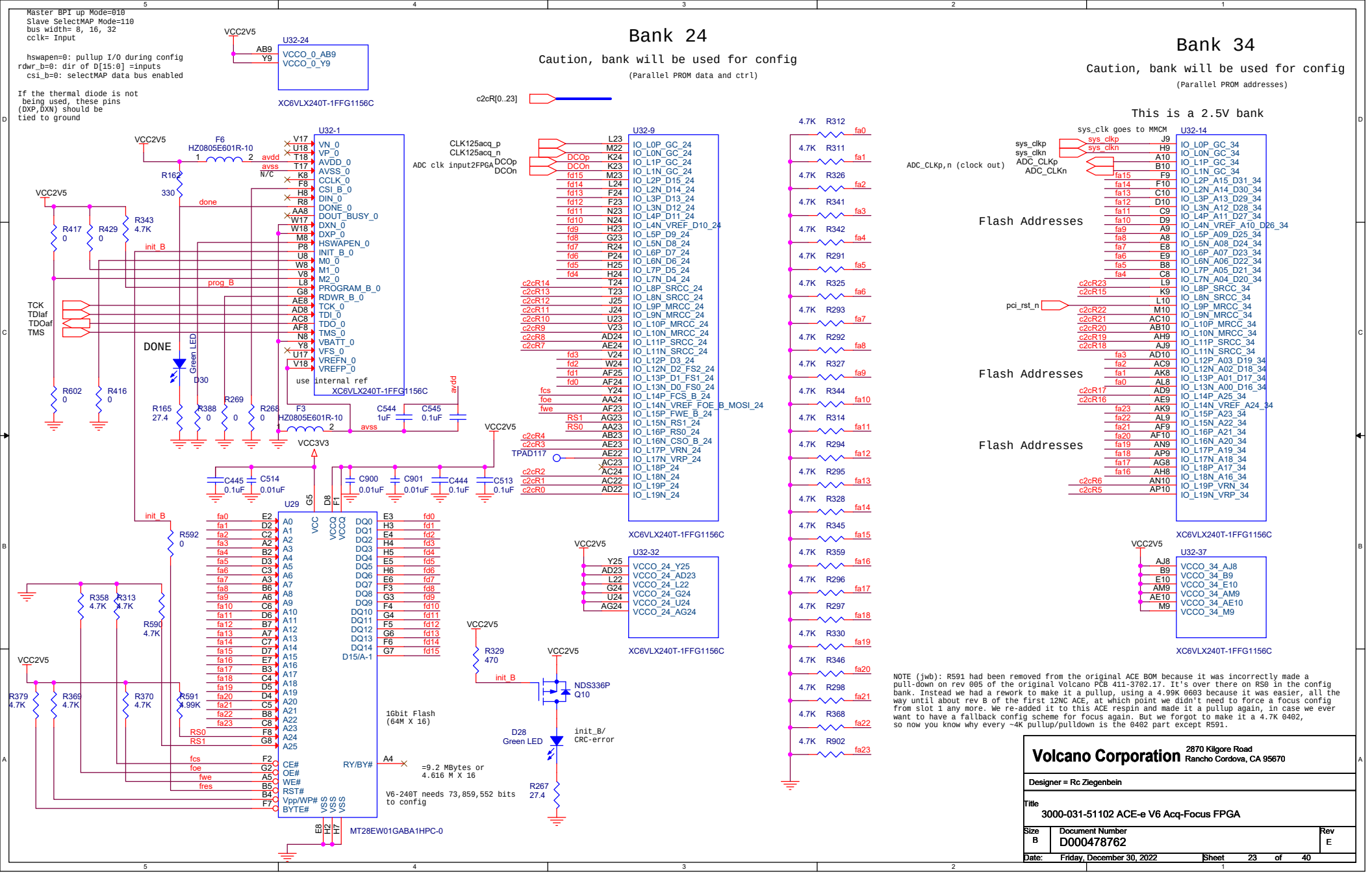
(Parallel PROM data and ctrl)

Bank 34

Caution, bank will be used for config

(Parallel PROM addresses)

This is a 2.5V bank



NOTE (jwb): R591 had been removed from the original ACE BOM because it was incorrectly made a pull-down on rev 005 of the original Volcano PCB 411-3702.17. It's over there on RS0 in the config bank. Instead we had a rework to make it a pullup, using a 4.99K 0603 because it was easier, all the way until about rev B of the first 12NC ACE, at which point we didn't need to force a focus config from slot 1 any more. We re-added it to this ACE respin and made it a pullup again, in case we ever want to have a fallback config scheme for focus again. But we forgot to make it a 4.7K 0402, so now you know why every ~4K pullup/pulldown is the 0402 part except R591.

Volcano Corporation 2870 Kilgore Road, Rancho Cordova, CA 95670

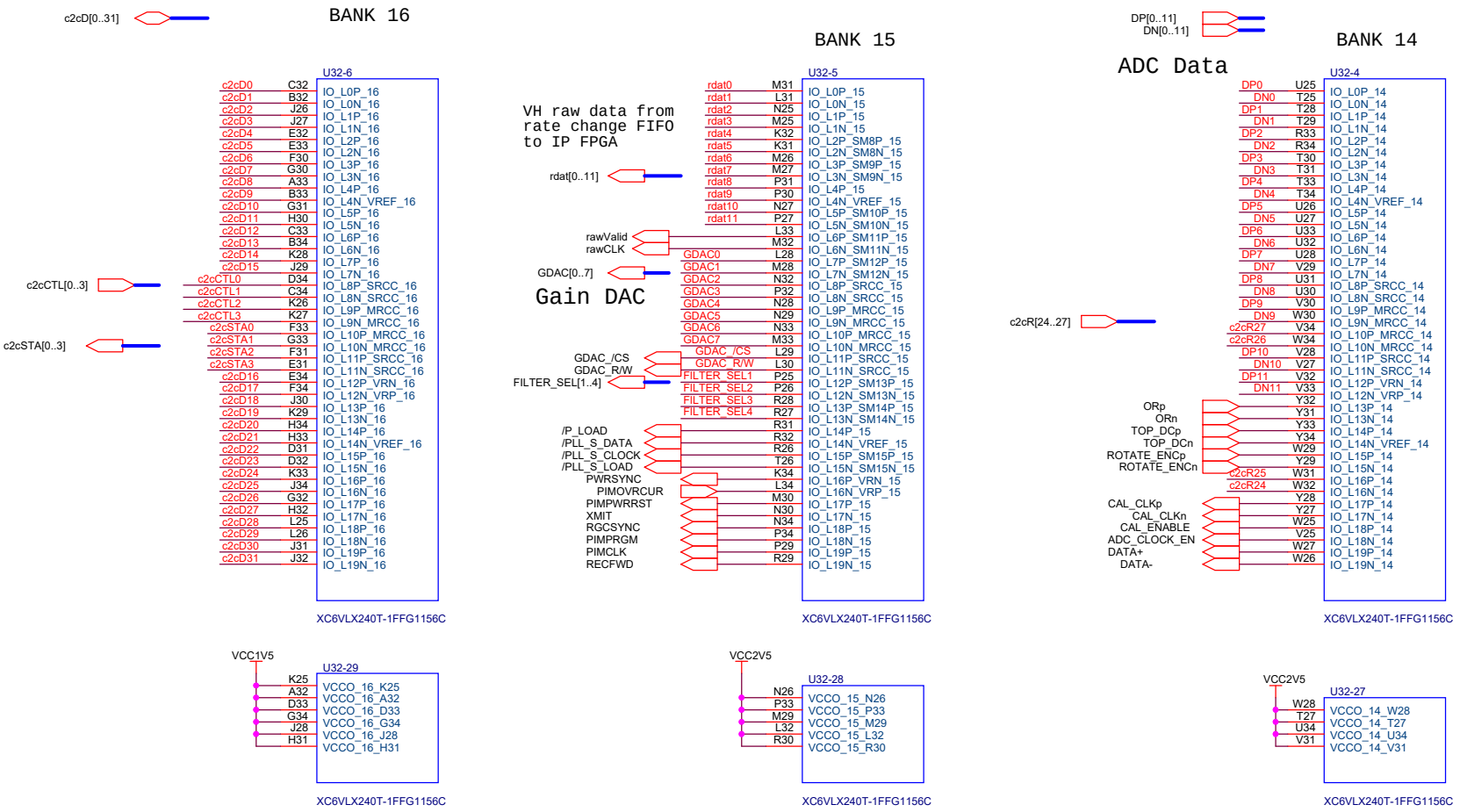
Designer = Rc Ziegenbein

Title: 3000-031-51102 ACE-e V6 Acq-Focus FPGA

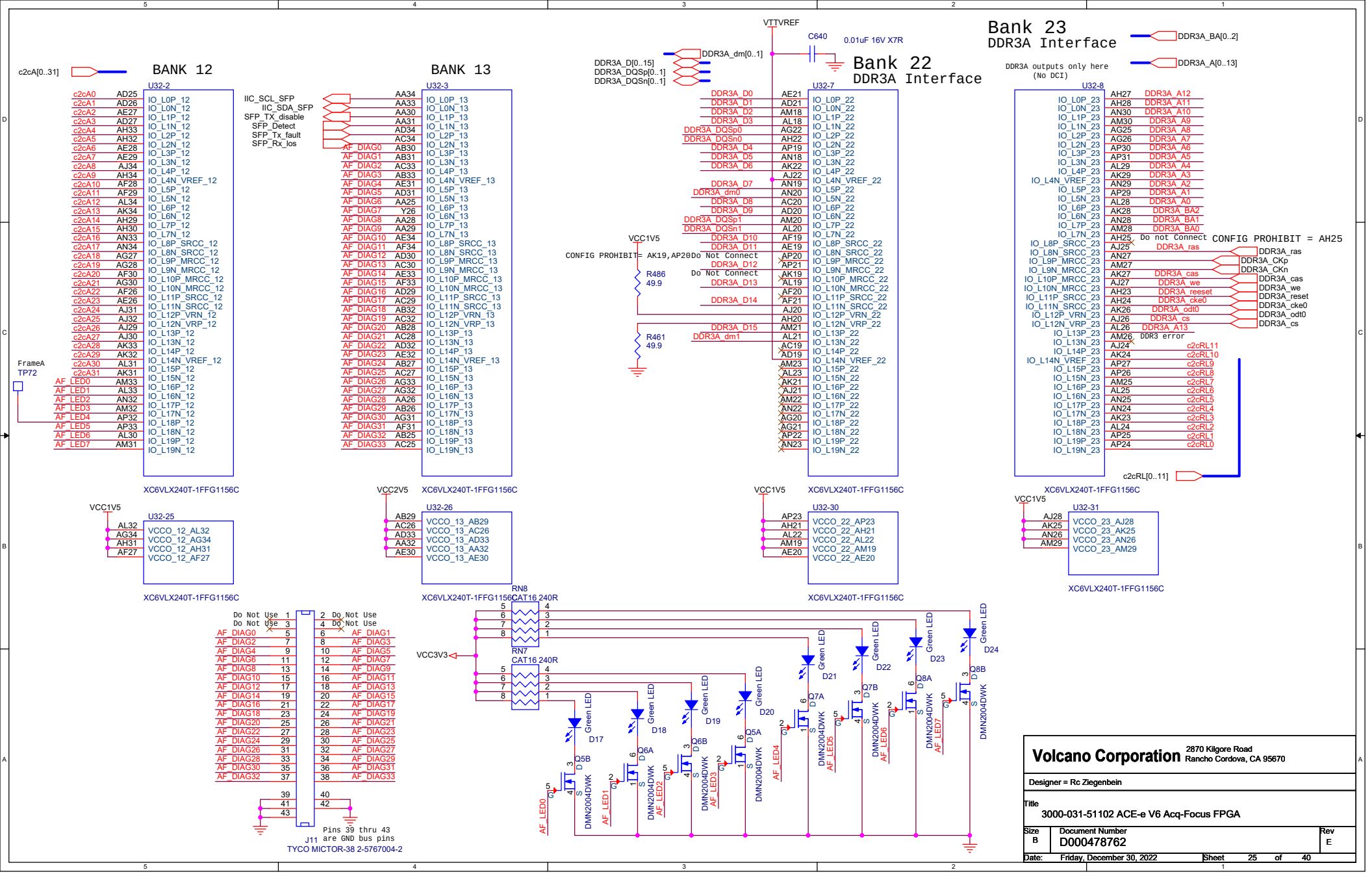
Size	Document Number	Rev
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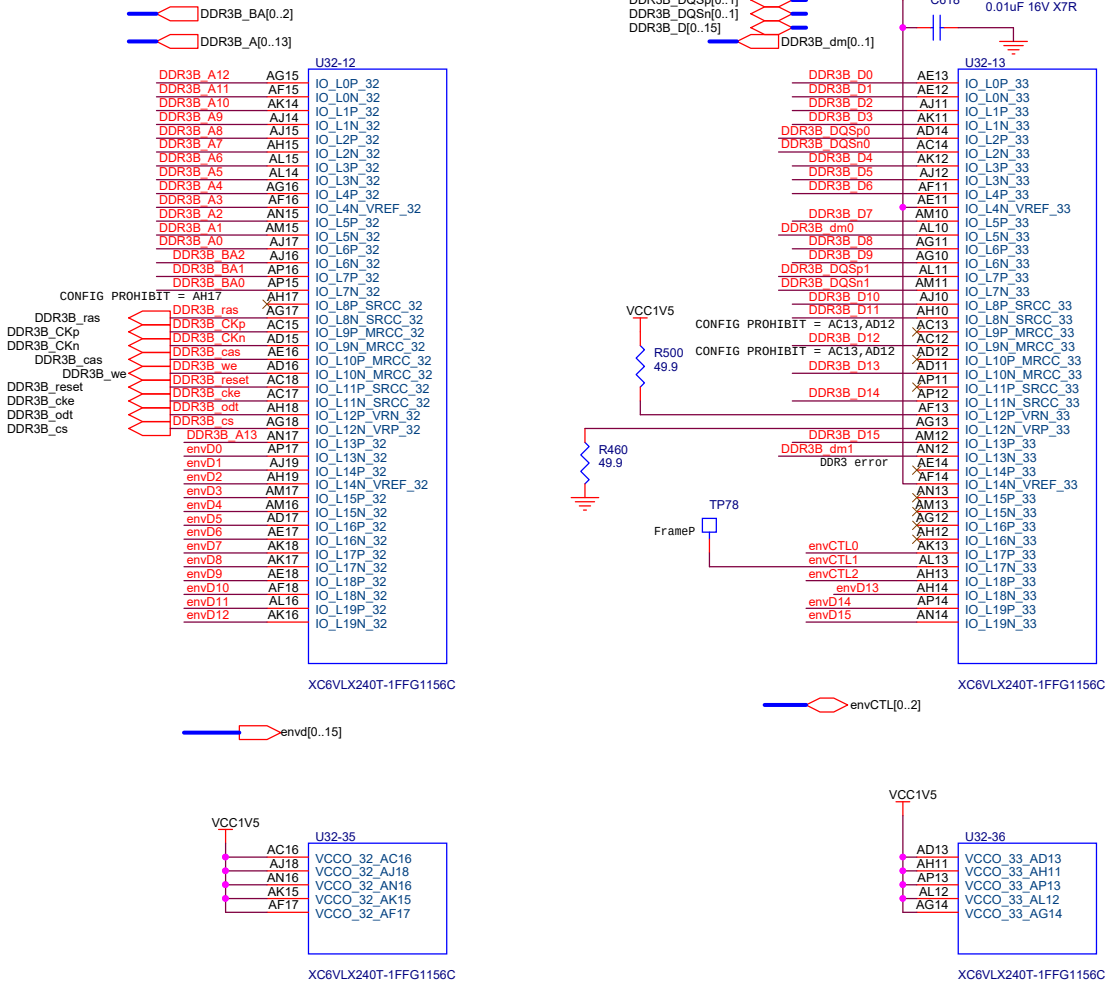
Analog Front End Signals



Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = Rc Ziegenbein		
Title: 3000-031-51102 ACE-e V6 Acq-Focus FPGA		
Size: B	Document Number: D000478762	Rev: E
Date: Friday, December 30, 2022	Sheet 24 of 40	

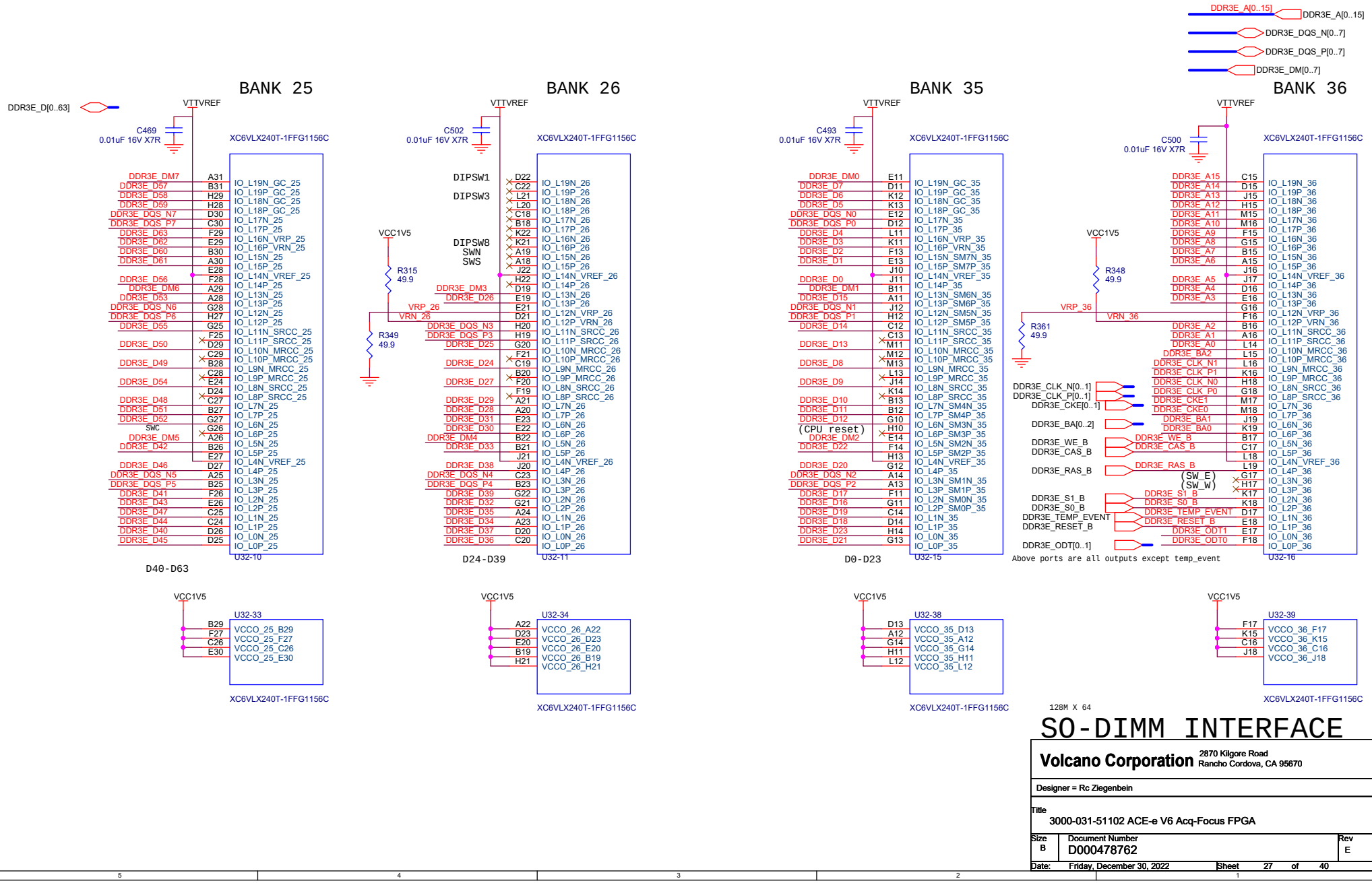

BANK 12
BANK 13
Bank 22
DDR3A Interface
Bank 23
DDR3A Interface
c2cA[0..31]
DDR3A_BA[0..2]
DDR3A_A[0..13]
DDR3A outputs only here
(No DCI)
DDR3A_D[0..15]
DDR3A_DQSp[0..1]
DDR3A_DQSn[0..1]
DDR3A_dm[0..1]
VTTVREF
C640
0.01uF 16V X7R
VCC1V5
VCC2V5
VCC3V3
CONFIG PROHIBIT= AK19,AP20
CONFIG PROHIBIT = AH25
Do Not Connect
R486
49.9
R461
49.9
FrameA
TP72
IIC_SCL_SFP
IIC_SDA_SFP
SFP_TX_disable
SFP_Detect
SFP_Tx_fault
SFP_Rx_los
U32-2
U32-3
U32-7
U32-8
U32-25
U32-26
U32-30
U32-31
XC6VLX240T-1FFG1156C
DDR3A_ras
DDR3A_CKp
DDR3A_CKn
DDR3A_cas
DDR3A_we
DDR3A_reset
DDR3A_cke0
DDR3A_odt0
DDR3A_cs
DDR3 error
c2cRL[0..11]
RN8
RN7
CAT16 240R
Green LED
DMN2004DWK
J11
TYCO MICTOR-38 2-5767004-2
Pins 39 thru 43
are GND bus pins
Do Not Use
Volcano Corporation
2870 Kilgore Road
Rancho Cordova, CA 95670
Designer = Rc Ziegenbein
Title
3000-031-51102 ACE-e V6 Acq-Focus FPGA
Size
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Document Number
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Bank 32
DDR3B interface

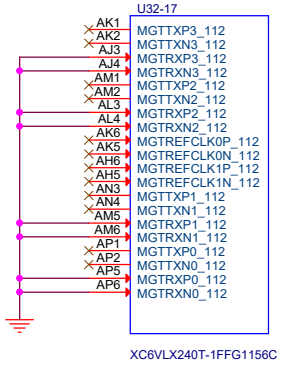


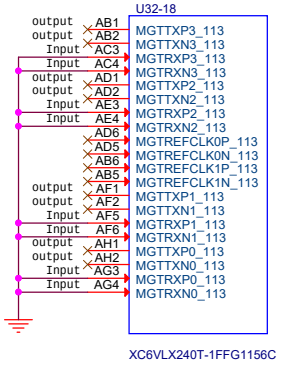
Bank 33
DDR3B interface

Volcano Corporation 2870 Kilgore Road, Rancho Cordova, CA 95670		
Designer = Rc Ziegenbein		
Title: 3000-031-51102 ACE-e V6 Acq-Focus FPGA		
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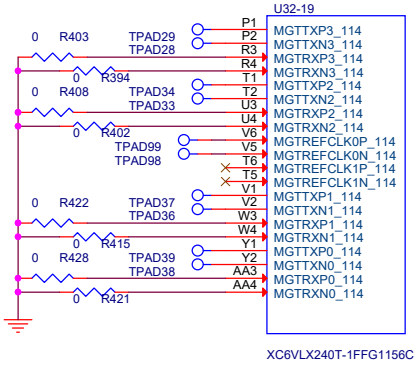

DDR3E_A[0..15]
DDR3E_DQS_N[0..7]
DDR3E_DQS_P[0..7]
DDR3E_DM[0..7]
BANK 25
BANK 26
BANK 35
BANK 36
DDR3E_D[0..63]
VTTVREF
C469
0.01uF 16V X7R
XC6VLX240T-1FFG1156C
D40-D63
U32-10
C502
DIPSW1
DIPSW3
DIPSW8
SWN
SWS
VCC1V5
R315
49.9
R349
VRP_26
VRN_26
D24-D39
U32-11
C493
D0-D23
U32-15
C500
R348
R361
VRP_36
VRN_36
DDR3E_CLK_N[0..1]
DDR3E_CLK_P[0..1]
DDR3E_CKE[0..1]
DDR3E_BA[0..2]
DDR3E_WE_B
DDR3E_CAS_B
DDR3E_RAS_B
DDR3E_S1_B
DDR3E_S0_B
DDR3E_TEMP_EVENT
DDR3E_RESET_B
DDR3E_ODT[0..1]
(CPU reset)
(SW_E)
(SW_W)
Above ports are all outputs except temp_event
U32-16
U32-33
U32-34
U32-38
U32-39
128M X 64
SO-DIMM INTERFACE
Volcano Corporation
2870 Kilgore Road
Rancho Cordova, CA 95670
Designer = Rc Ziegenbein
Title
3000-031-51102 ACE-e V6 Acq-Focus FPGA
Size
B
Document Number
D000478762
Rev
E
Date: Friday, December 30, 2022
Sheet 27 of 40

To/From PCIe Switch port 5, (4 lanes)

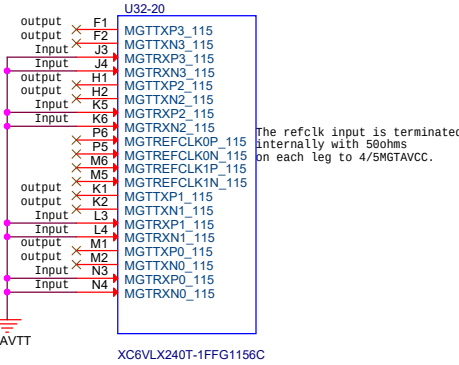




The refclk input is terminated internally with 50ohms on each leg to 4/5MGTAVCC.



To/From PCIe Switch port 4, (4 lanes)



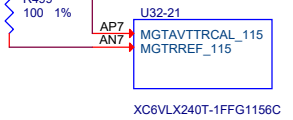
The refclk input is terminated internally with 50ohms on each leg to 4/5MGTAVCC.

MGT_AVTT

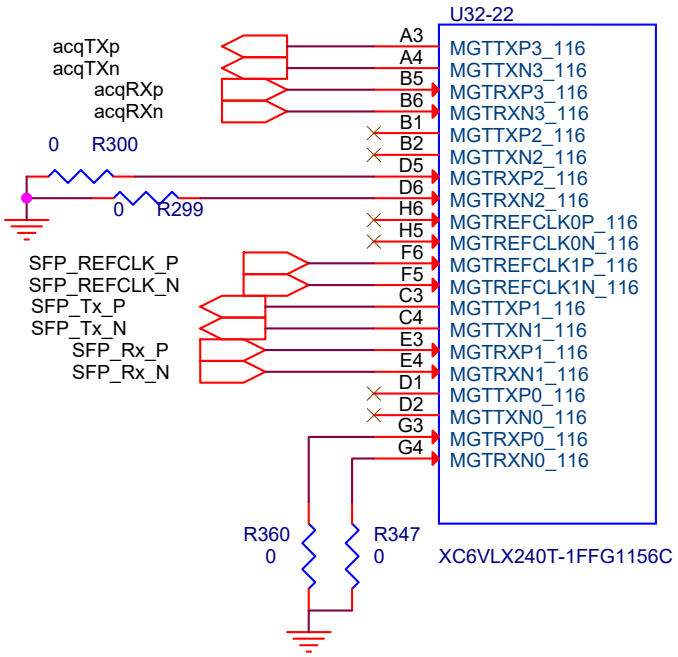
This voltage is 1.2V

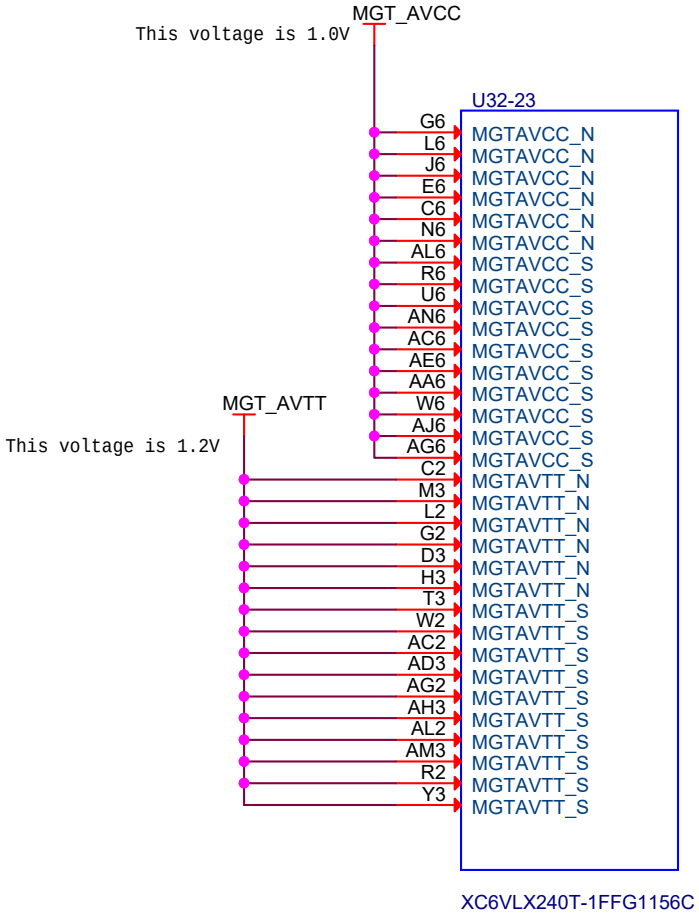
R499 100 1%

Precision reference resistor for internal calibration termination



Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = Rc Ziegenbein		
Title 3000-031-51102 ACE-e V6 Acq-Focus FPGA		
Size B	Document Number D000478762	Rev E
Date: Friday, December 30, 2022	Sheet 28 of 40	





Volcano Corporation 2870 Kilgore Road, Rancho Cordova, CA 95670

Designer = Rc Ziegenbein

Title: 3000-031-51102 ACE-e V6 Acq-Focus FPGA

Size	Document Number	Rev
A	D000478762	E

Date: Friday, December 30, 2022 Sheet 29 of 40

This voltage is 1.0V

VCCINT

U32-40

V19	VCCINT_V19
W12	VCCINT_W12
R14	VCCINT_R14
T15	VCCINT_T15
T19	VCCINT_T19
AA20	VCCINT_AA20
N18	VCCINT_N18
P11	VCCINT_P11
P15	VCCINT_P15
P19	VCCINT_P19
V21	VCCINT_V21
T21	VCCINT_T21
U10	VCCINT_U10
U12	VCCINT_U12
U14	VCCINT_U14
U16	VCCINT_U16
U20	VCCINT_U20
R12	VCCINT_R12
T11	VCCINT_T11
V15	VCCINT_V15
R10	VCCINT_R10
R20	VCCINT_R20
T13	VCCINT_T13
R16	VCCINT_R16
AA12	VCCINT_AA12
R18	VCCINT_R18
AA16	VCCINT_AA16
AA14	VCCINT_AA14
W16	VCCINT_W16
AA18	VCCINT_AA18
W20	VCCINT_W20
AB11	VCCINT_AB11
N16	VCCINT_N16
N20	VCCINT_N20
AB13	VCCINT_AB13
P13	VCCINT_P13
AB17	VCCINT_AB17
P17	VCCINT_P17
AB21	VCCINT_AB21
P21	VCCINT_P21
M21	VCCINT_M21
N14	VCCINT_N14
Y11	VCCINT_Y11
AB15	VCCINT_AB15
Y13	VCCINT_Y13
AB19	VCCINT_AB19
Y15	VCCINT_Y15
M19	VCCINT_M19
Y17	VCCINT_Y17
N12	VCCINT_N12
Y19	VCCINT_Y19
Y21	VCCINT_Y21
V11	VCCINT_V11
V13	VCCINT_V13
W14	VCCINT_W14

XC6VLX240T-1FFG1156C

This voltage is 2.5V

VCCAUX

U32-41

N22	VCCAUX_N22
P23	VCCAUX_P23
T9	VCCAUX_T9
V9	VCCAUX_V9
W10	VCCAUX_W10
P9	VCCAUX_P9
Y23	VCCAUX_Y23
R22	VCCAUX_R22
AA10	VCCAUX_AA10
U22	VCCAUX_U22
AA22	VCCAUX_AA22
N10	VCCAUX_N10
W22	VCCAUX_W22

XC6VLX240T-1FFG1156C

U32-42

U13	GND_U13
R25	GND_R25
F7	GND_F7
Y14	GND_Y14
W5	GND_W5
J7	GND_J7
G29	GND_G29
M4	GND_M4
K10	GND_K10
N5	GND_N5
P16	GND_P16
N21	GND_N21
A1	GND_A1
R9	GND_R9
A2	GND_A2
T22	GND_T22
A5	GND_A5
V10	GND_V10
A6	GND_A6
W23	GND_W23
A7	GND_A7
E15	GND_E15
A17	GND_A17
G1	GND_G1
A27	GND_A27
H26	GND_H26
A34	GND_A34
J33	GND_J33
AA1	GND_AA1
L5	GND_L5
AA2	GND_AA2
M24	GND_M24
AA5	GND_AA5
N13	GND_N13
AA7	GND_AA7
P7	GND_P7
AA9	GND_AA9
P28	GND_P28
AA11	GND_AA11
R17	GND_R17
AA13	GND_AA13
T12	GND_T12
AA15	GND_AA15
U5	GND_U5
AA17	GND_AA17
U29	GND_U29
AA19	GND_AA19
V20	GND_V20
AA21	GND_AA21
W13	GND_W13

XC6VLX240T-1FFG1156C

U32-43

AA27	GND_AA27
Y6	GND_Y6
AB3	GND_AB3
Y22	GND_Y22
AB4	GND_AB4
F3	GND_F3
AB7	GND_AB7
F22	GND_F22
AB8	GND_AB8
G7	GND_G7
AB12	GND_AB12
H7	GND_H7
AB14	GND_AB14
J2	GND_J2
AB16	GND_AB16
J13	GND_J13
AB18	GND_AB18
K4	GND_K4
AB20	GND_AB20
K30	GND_K30
AB22	GND_AB22
L17	GND_L17
AB24	GND_AB24
M14	GND_M14
AB34	GND_AB34
N1	GND_N1
AC1	GND_AC1
N9	GND_N9
AC5	GND_AC5
N17	GND_N17
AC7	GND_AC7
P3	GND_P3
AC11	GND_AC11
P12	GND_P12
AC21	GND_AC21
P20	GND_P20
AC31	GND_AC31
R5	GND_R5
AD4	GND_AD4
R13	GND_R13
AD7	GND_AD7
R21	GND_R21
AD18	GND_AD18
T7	GND_T7
AD28	GND_AD28
T16	GND_T16
AE1	GND_AE1
U1	GND_U1
AE2	GND_AE2
U9	GND_U9

XC6VLX240T-1FFG1156C

U32-44

AE5	GND_AE5
U19	GND_U19
AE7	GND_AE7
V4	GND_V4
AE15	GND_AE15
V14	GND_V14
AE25	GND_AE25
V26	GND_V26
AF3	GND_AF3
W9	GND_W9
AF4	GND_AF4
W19	GND_W19
AF7	GND_AF7
Y4	GND_Y4
AF12	GND_AF12
Y10	GND_Y10
AF22	GND_AF22
Y18	GND_Y18
AF32	GND_AF32
T8	GND_T8
AG1	GND_AG1
E25	GND_E25
AG5	GND_AG5
F4	GND_F4
AG7	GND_AG7
F12	GND_F12
AG9	GND_AG9
F32	GND_F32
AG19	GND_AG19
G5	GND_G5
AG29	GND_AG29
G19	GND_G19
AH4	GND_AH4
H4	GND_H4
AH7	GND_AH7
H16	GND_H16
AH16	GND_AH16
J1	GND_J1
AH26	GND_AH26
J5	GND_J5
AJ1	GND_AJ1
J8	GND_J8
AJ2	GND_AJ2
J23	GND_J23
AJ5	GND_AJ5
K3	GND_K3
AJ7	GND_AJ7
K7	GND_K7
AJ13	GND_AJ13
K20	GND_K20

XC6VLX240T-1FFG1156C

U32-45

AJ23	GND_AJ23
L1	GND_L1
AJ33	GND_AJ33
L7	GND_L7
AK3	GND_AK3
L27	GND_L27
AK4	GND_AK4
M7	GND_M7
AK7	GND_AK7
M20	GND_M20
AK10	GND_AK10
M34	GND_M34
AK20	GND_AK20
N2	GND_N2
AK30	GND_AK30
N7	GND_N7
AL1	GND_AL1
N11	GND_N11
AL5	GND_AL5
N15	GND_N15
AL7	GND_AL7
N19	GND_N19
AL17	GND_AL17
N31	GND_N31
AL27	GND_AL27
P4	GND_P4
AM4	GND_AM4
P10	GND_P10
AM7	GND_AM7
P14	GND_P14
AM8	GND_AM8
P18	GND_P18
AM14	GND_AM14
P22	GND_P22
AM24	GND_AM24
R1	GND_R1
AM34	GND_AM34
R7	GND_R7
AN1	GND_AN1
R11	GND_R11
AN2	GND_AN2
R15	GND_R15
AN5	GND_AN5
R19	GND_R19
AN8	GND_AN8
R23	GND_R23
AN11	GND_AN11
T4	GND_T4
AN21	GND_AN21
T10	GND_T10

XC6VLX240T-1FFG1156C

U32-46

AN31	GND_AN31
T14	GND_T14
AP3	GND_AP3
T20	GND_T20
AP4	GND_AP4
T32	GND_T32
AP8	GND_AP8
U2	GND_U2
AP18	GND_AP18
U7	GND_U7
AP28	GND_AP28
U11	GND_U11
AP34	GND_AP34
U15	GND_U15
B3	GND_B3
U21	GND_U21
B4	GND_B4
V3	GND_V3
B7	GND_B7
V7	GND_V7
B14	GND_B14
V12	GND_V12
B24	GND_B24
V16	GND_V16
C1	GND_C1
V22	GND_V22
C5	GND_C5
W1	GND_W1
C7	GND_C7
W7	GND_W7
C11	GND_C11
W11	GND_W11
C21	GND_C21
W15	GND_W15
C31	GND_C31
W21	GND_W21
D4	GND_D4
W33	GND_W33
D7	GND_D7
Y5	GND_Y5
D8	GND_D8
Y7	GND_Y7
D18	GND_D18
Y12	GND_Y12
D28	GND_D28
Y16	GND_Y16
E1	GND_E1
Y20	GND_Y20
E2	GND_E2
Y30	GND_Y30

XC6VLX240T-1FFG1156C

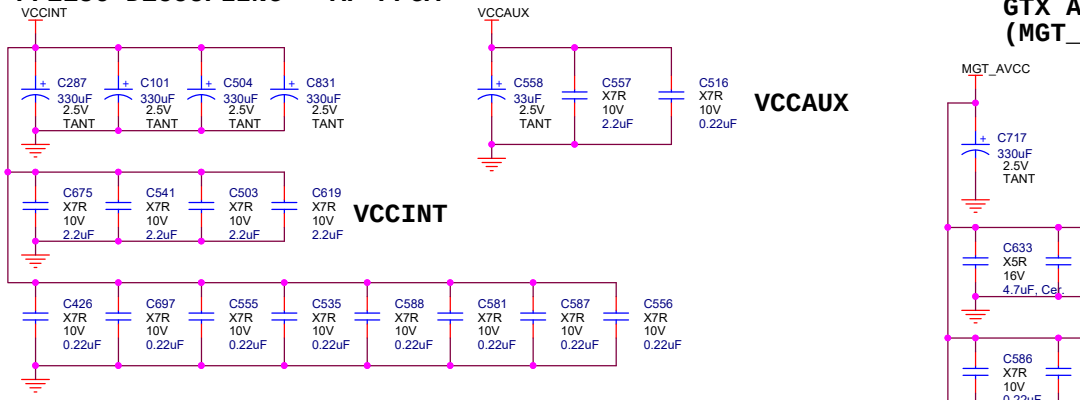
U32-47

E5	GND_E5
E7	GND_E7
G9	GND_G9

XC6VLX240T-1FFG1156C

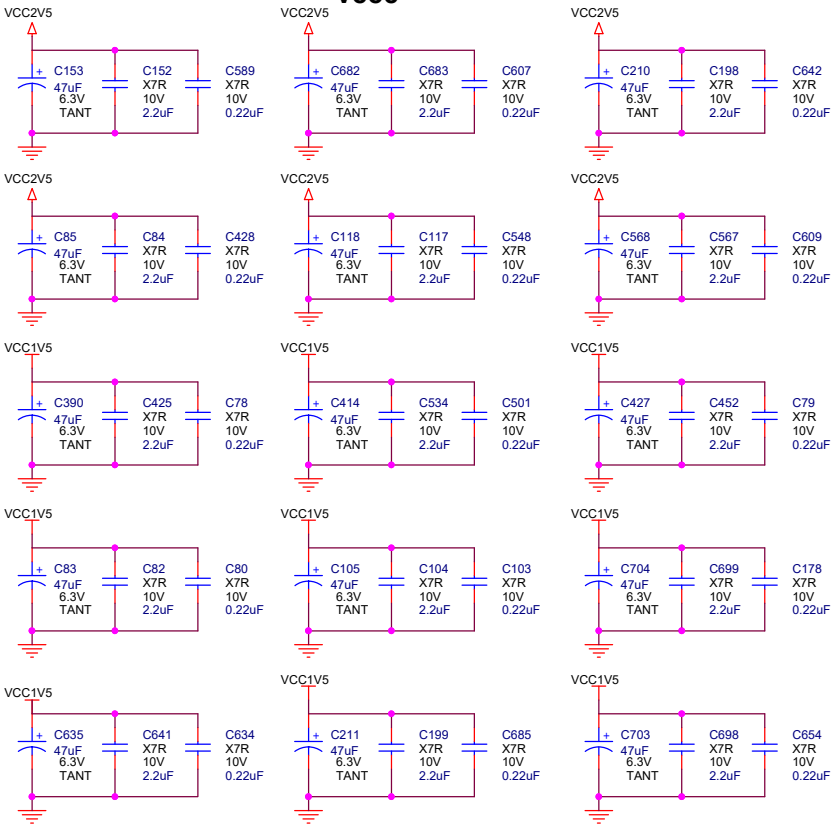
Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = Rc Ziegenbein		
Title 3000-031-51102 ACE-e V6 Acq-Focus FPGA		
Size B	Document Number D000478762	Rev E
Date: Friday, December 30, 2022	Sheet 30 of 40	

V-6 FF1156 DECOUPLING - AF FPGA



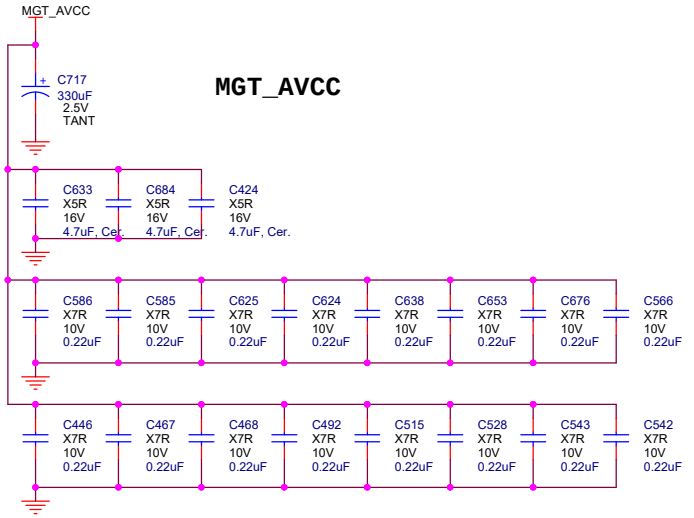
VCCO

One module per bank: 6 banks use VCC2V5, 10 use VCC1V5

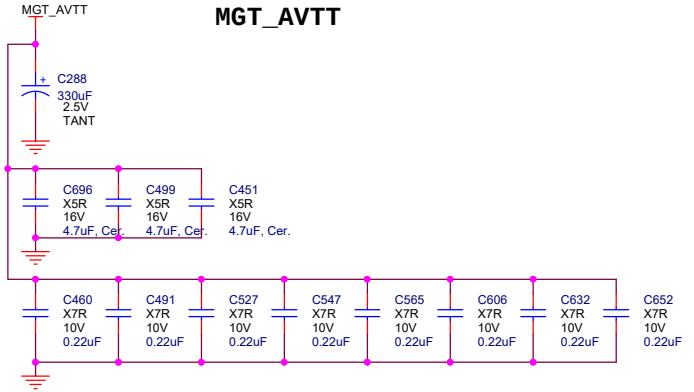


GTX ANALOG VOLTAGE SUPPLIES (MGT_AVCC, MGT_AVTT)

MGT_AVCC



MGT_AVTT



PLACEMENT RULES

330uF, 100uF, 47uF, 33uF: Anywhere on board, but as close as possible to FPGA.

4.7uF, 2.2uF: No more than 3" from periphery of FPGA, but as close as possible.

0.22uF: No more than 1" from periphery of FPGA, but as close as possible.

Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = M. Shell		
Title: 3000-031-51102 ACE-e V6 Acq-Focus FPGA Bypass		
Size: B	Document Number: D000478762	Rev: E
Date: Friday, December 30, 2022	Sheet 31 of 40	

Slave SelectMAP Mode=110 Master BPI up Mode=010
bus width= 8, 16, 32
cclk= Input

hswapen=0: pullup I/O during config
rdwr_b=0: dir of D[15:0] =inputs
csi_b=0: selectMAP data bus enabled

If the thermal diode is not being used, these pins (DXP,DXN) should be tied to ground

Bank 24

Caution, bank will be used for config

(Parallel PROM data and ctrl)

Bank 34

Caution, bank will be used for config

(Parallel PROM addresses)

U31-24 VCCO_0_AB9 VCCO_0_Y9 — XC6VLX240T-1FFG1156C

U31-1 — XC6VLX240T-1FFG1156C

U31-9 — XC6VLX240T-1FFG1156C

U31-14 — XC6VLX240T-1FFG1156C

U31-32 — XC6VLX240T-1FFG1156C

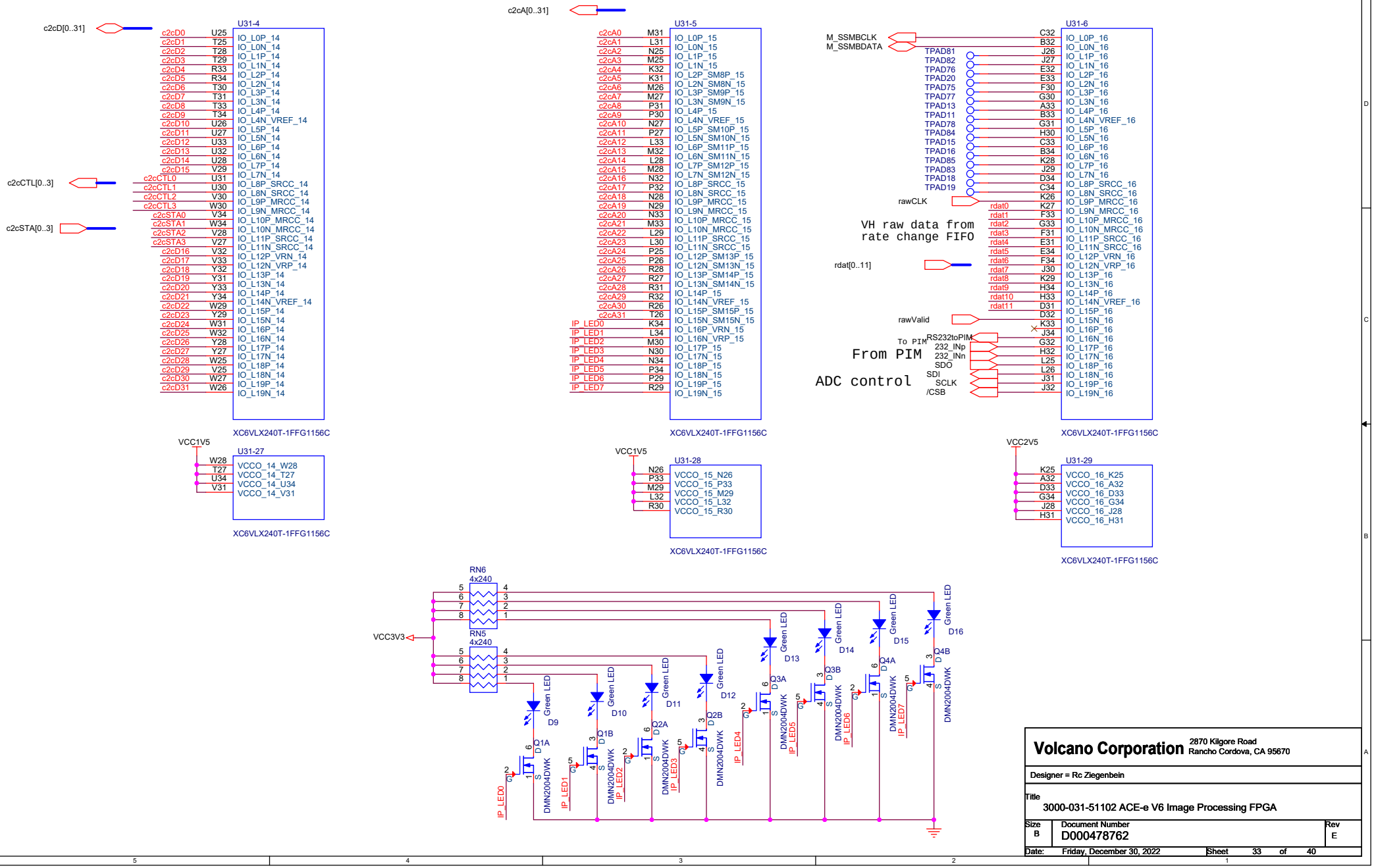
U31-37 — XC6VLX240T-1FFG1156C

U28 — MT28EW01GABA1HPC-0 (1Gbit Flash (64M X 16))

V6-240T needs 73,859,552 bits to config
=9.2 MBytes or 4.616 M X 16

J10 TYCO MICTOR-38 2-5767004-2 — Pins 39 thru 43 are GND bus pins

Volcano Corporation	2870 Kilgore Road, Rancho Cordova, CA 95670
Designer = Rc Ziegenbein	
Title	3000-031-51102 ACE-e V6 Image Processing FPGA
Size B	Document Number D000478762 Rev E
Date: Friday, December 30, 2022	Sheet 32 of 40



Volcano Corporation 2870 Kilgore Road, Rancho Cordova, CA 95670

Designer = Rc Ziegenbein

Title: 3000-031-51102 ACE-e V6 Image Processing FPGA

Size	Document Number	Rev
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Date: Friday, December 30, 2022 Sheet 33 of 40

Bank 12

ODD1_D[0..15]
ODD1_A[0..19]
ODD1_CTRL[0..2]
AoddD[] = ODD1_D[0..15]

U31-2

ODD1_D0	AD25	IO_L0P_12
ODD1_D1	AD26	IO_L0N_12
ODD1_D2	AE27	IO_L1P_12
ODD1_D3	AD27	IO_L1N_12
ODD1_D4	AH33	IO_L2P_12
ODD1_D5	AH32	IO_L2N_12
ODD1_D6	AE28	IO_L3P_12
ODD1_D7	AE29	IO_L3N_12
ODD1_D8	AJ34	IO_L4P_12
	AH34	IO_L4N_VREF_12
ODD1_D9	AF28	IO_L5P_12
ODD1_D10	AF29	IO_L5N_12
ODD1_D11	AL34	IO_L6P_12
ODD1_D12	AK34	IO_L6N_12
ODD1_D13	AH29	IO_L7P_12
ODD1_D14	AH30	IO_L7N_12
ODD1_D15	AN33	IO_L8P_SRCC_12
ODD1_A0	AN34	IO_L8N_SRCC_12
ODD1_A1	AG27	IO_L9P_MRCC_12
ODD1_A2	AG28	IO_L9N_MRCC_12
ODD1_A3	AF30	IO_L10P_MRCC_12
ODD1_A4	AG30	IO_L10N_MRCC_12
ODD1_A5	AF26	IO_L11P_SRCC_12
ODD1_A6	AE26	IO_L11N_SRCC_12
ODD1_A7	AJ31	IO_L12P_VRN_12
ODD1_A8	AJ32	IO_L12N_VRP_12
ODD1_A9	AJ29	IO_L13P_12
ODD1_A10	AJ30	IO_L13N_12
ODD1_A11	AK33	IO_L14P_12
ODD1_A12	AK32	IO_L14N_VREF_12
ODD1_A13	AL31	IO_L15P_12
ODD1_A14	AK31	IO_L15N_12
ODD1_A15	AM33	IO_L16P_12
ODD1_A16	AL33	IO_L16N_12
ODD1_A17	AN32	IO_L17P_12
ODD1_A18	AM32	IO_L17N_12
ODD1_A19	AP32	IO_L18P_12
ODD1_CTRL0	AP33	IO_L18N_12
ODD1_CTRL1	AL30	IO_L19P_12
ODD1_CTRL2	AM31	IO_L19N_12

XC6VLX240T-1FFG1156C

VCC2V5

U31-25

AL32	VCCO_12_AL32
AG34	VCCO_12_AG34
AH31	VCCO_12_AH31
AF27	VCCO_12_AF27

XC6VLX240T-1FFG1156C

Bank 13

EVEN1_D[0..15]
EVEN1_A[0..19]
EVEN1_CTRL[0..2]

U31-3

EVEN1_D0	AA34	IO_L0P_13
EVEN1_D1	AA33	IO_L0N_13
EVEN1_D2	AA30	IO_L1P_13
EVEN1_D3	AA31	IO_L1N_13
EVEN1_D4	AD34	IO_L2P_13
EVEN1_D5	AC34	IO_L2N_13
EVEN1_D6	AB30	IO_L3P_13
EVEN1_D7	AB31	IO_L3N_13
EVEN1_D8	AC33	IO_L4P_13
	AB33	IO_L4N_VREF_13
EVEN1_D9	AE31	IO_L5P_13
EVEN1_D10	AD31	IO_L5N_13
EVEN1_D11	AA25	IO_L6P_13
EVEN1_D12	Y26	IO_L6N_13
EVEN1_D13	AA28	IO_L7P_13
EVEN1_D14	AA29	IO_L7N_13
EVEN1_D15	AE34	IO_L8P_SRCC_13
EVEN1_A0	AF34	IO_L8N_SRCC_13
EVEN1_A1	AD30	IO_L9P_MRCC_13
EVEN1_A2	AC30	IO_L9N_MRCC_13
EVEN1_A3	AE33	IO_L10P_MRCC_13
EVEN1_A4	AF33	IO_L10N_MRCC_13
EVEN1_A5	AD29	IO_L11P_SRCC_13
EVEN1_A6	AC29	IO_L11N_SRCC_13
EVEN1_A7	AB32	IO_L12P_VRN_13
EVEN1_A8	AC32	IO_L12N_VRP_13
EVEN1_A9	AB28	IO_L13P_13
EVEN1_A10	AC28	IO_L13N_13
EVEN1_A11	AD32	IO_L14P_13
EVEN1_A12	AE32	IO_L14N_VREF_13
EVEN1_A13	AB27	IO_L15P_13
EVEN1_A14	AC27	IO_L15N_13
EVEN1_A15	AG33	IO_L16P_13
EVEN1_A16	AG32	IO_L16N_13
EVEN1_A17	AA26	IO_L17P_13
EVEN1_A18	AB26	IO_L17N_13
EVEN1_A19	AG31	IO_L18P_13
EVEN1_CTRL0	AF31	IO_L18N_13
EVEN1_CTRL1	AB25	IO_L19P_13
EVEN1_CTRL2	AC25	IO_L19N_13

XC6VLX240T-1FFG1156C

VCC2V5

U31-26

AB29	VCCO_13_AB29
AC26	VCCO_13_AC26
AD33	VCCO_13_AD33
AA32	VCCO_13_AA32
AE30	VCCO_13_AE30

XC6VLX240T-1FFG1156C

Bank 22

ODD2_D[0..15]
ODD2_A[0..19]
ODD2_CTRL[0..2]
BoddD[] = ODD2_D[0..15]

U31-7

ODD2_D0	AE21	IO_L0P_22
ODD2_D1	AD21	IO_L0N_22
ODD2_D2	AM18	IO_L1P_22
ODD2_D3	AL18	IO_L1N_22
ODD2_D4	AG22	IO_L2P_22
ODD2_D5	AH22	IO_L2N_22
ODD2_D6	AP19	IO_L3P_22
ODD2_D7	AN18	IO_L3N_22
ODD2_D8	AK22	IO_L4P_22
	AJ22	IO_L4N_VREF_22
ODD2_D9	AN19	IO_L5P_22
ODD2_D10	AN20	IO_L5N_22
ODD2_D11	AC20	IO_L6P_22
ODD2_D12	AD20	IO_L6N_22
ODD2_D13	AM20	IO_L7P_22
ODD2_D14	AL20	IO_L7N_22
ODD2_D15	AF19	IO_L8P_SRCC_22
ODD2_A0	AE19	IO_L8N_SRCC_22
ODD2_A1	AP20	IO_L9P_MRCC_22
ODD2_A2	AP21	IO_L9N_MRCC_22
ODD2_A3	AK19	IO_L10P_MRCC_22
ODD2_A4	AL19	IO_L10N_MRCC_22
ODD2_A5	AF20	IO_L11P_SRCC_22
ODD2_A6	AF21	IO_L11N_SRCC_22
ODD2_A7	AJ20	IO_L12P_VRN_22
ODD2_A8	AH20	IO_L12N_VRP_22
ODD2_A9	AM21	IO_L13P_22
ODD2_A10	AL21	IO_L13N_22
ODD2_A11	AC19	IO_L14P_22
ODD2_A12	AD19	IO_L14N_VREF_22
ODD2_A13	AM23	IO_L15P_22
ODD2_A14	AL23	IO_L15N_22
ODD2_A15	AK21	IO_L16P_22
ODD2_A16	AJ21	IO_L16N_22
ODD2_A17	AM22	IO_L17P_22
ODD2_A18	AN22	IO_L17N_22
ODD2_A19	AG20	IO_L18P_22
ODD2_CTRL0	AG21	IO_L18N_22
ODD2_CTRL1	AP22	IO_L19P_22
ODD2_CTRL2	AN23	IO_L19N_22

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VCC2V5

U31-30

AP23	VCCO_22_AP23
AH21	VCCO_22_AH21
AL22	VCCO_22_AL22
AM19	VCCO_22_AM19
AE20	VCCO_22_AE20

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Bank 23

EVEN2_D[0..15]
EVEN2_A[0..19]
EVEN2_CTRL[0..2]

U31-8

EVEN2_D0	AH27	IO_L0P_23
EVEN2_D1	AH28	IO_L0N_23
EVEN2_D2	AN30	IO_L1P_23
EVEN2_D3	AM30	IO_L1N_23
EVEN2_D4	AG25	IO_L2P_23
EVEN2_D5	AG26	IO_L2N_23
EVEN2_D6	AP30	IO_L3P_23
EVEN2_D7	AP31	IO_L3N_23
EVEN2_D8	AL29	IO_L4P_23
	AK29	IO_L4N_VREF_23
EVEN2_D9	AN29	IO_L5P_23
EVEN2_D10	AP29	IO_L5N_23
EVEN2_D11	AL28	IO_L6P_23
EVEN2_D12	AK28	IO_L6N_23
EVEN2_D13	AN28	IO_L7P_23
EVEN2_D14	AM28	IO_L7N_23
EVEN2_D15	AH25	IO_L8P_SRCC_23
EVEN2_A0	AJ25	IO_L8N_SRCC_23
EVEN2_A1	AN27	IO_L9P_MRCC_23
EVEN2_A2	AM27	IO_L9N_MRCC_23
EVEN2_A3	AK27	IO_L10P_MRCC_23
EVEN2_A4	AJ27	IO_L10N_MRCC_23
EVEN2_A5	AH23	IO_L11P_SRCC_23
EVEN2_A6	AH24	IO_L11N_SRCC_23
EVEN2_A7	AK26	IO_L12P_VRN_23
EVEN2_A8	AJ26	IO_L12N_VRP_23
EVEN2_A9	AL26	IO_L13P_23
EVEN2_A10	AM26	IO_L13N_23
EVEN2_A11	AJ24	IO_L14P_23
EVEN2_A12	AK24	IO_L14N_VREF_23
EVEN2_A13	AP27	IO_L15P_23
EVEN2_A14	AP26	IO_L15N_23
EVEN2_A15	AM25	IO_L16P_23
EVEN2_A16	AL25	IO_L16N_23
EVEN2_A17	AN25	IO_L17P_23
EVEN2_A18	AN24	IO_L17N_23
EVEN2_A19	AK23	IO_L18P_23
EVEN2_CTRL0	AL24	IO_L18N_23
EVEN2_CTRL1	AP25	IO_L19P_23
EVEN2_CTRL2	AP24	IO_L19N_23

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VCC2V5

U31-31

AJ28	VCCO_23_AJ28
AK25	VCCO_23_AK25
AN26	VCCO_23_AN26
AM29	VCCO_23_AM29

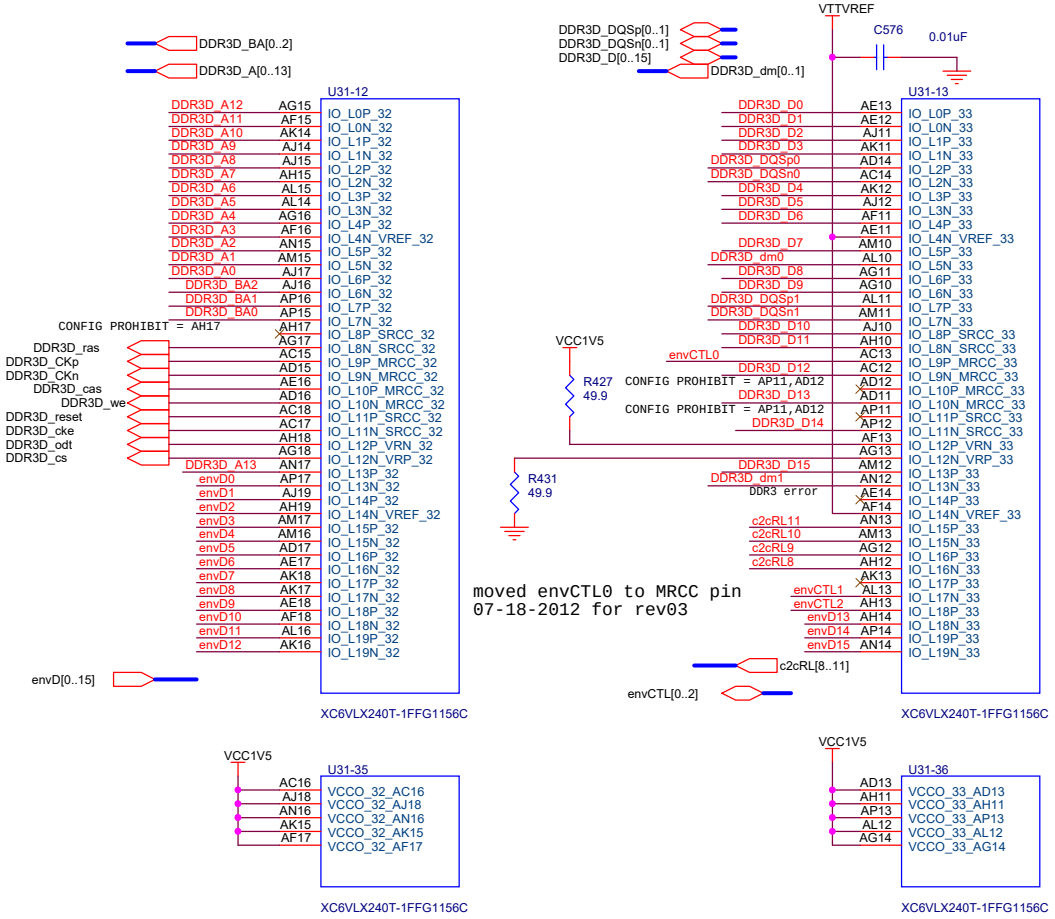
XC6VLX240T-1FFG1156C

SRAM (Scan input) Interface

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Bank 32
DDR3 D interface

Bank 33
DDR3 D interface



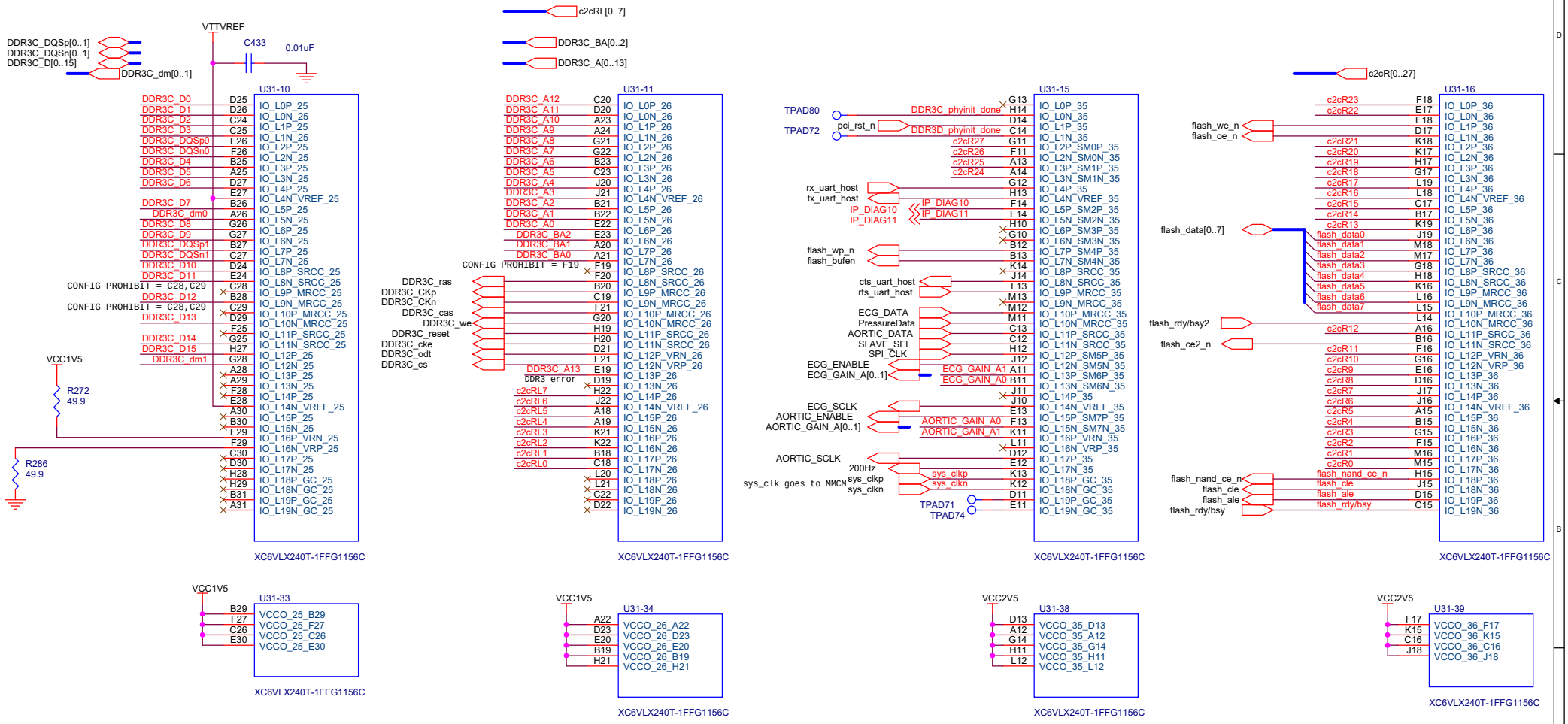
Volcano Corporation	2870 Kilgore Road Rancho Cordova, CA 95670	
Designer = Rc Ziegenbein		
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Bank 25
DDR3 C interface

Bank 26
DDR3 C interface

Bank 35
Misc. interfaces

Bank 36
Flash interfaces



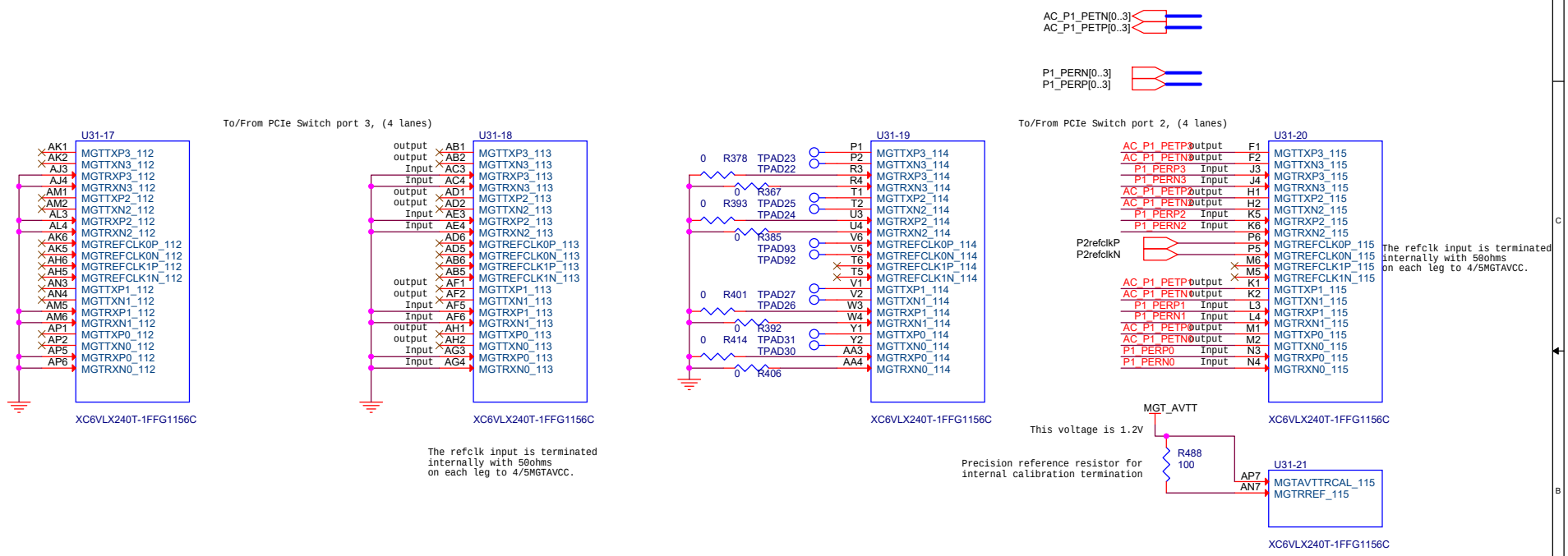
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Designer = Rc Ziegenbein

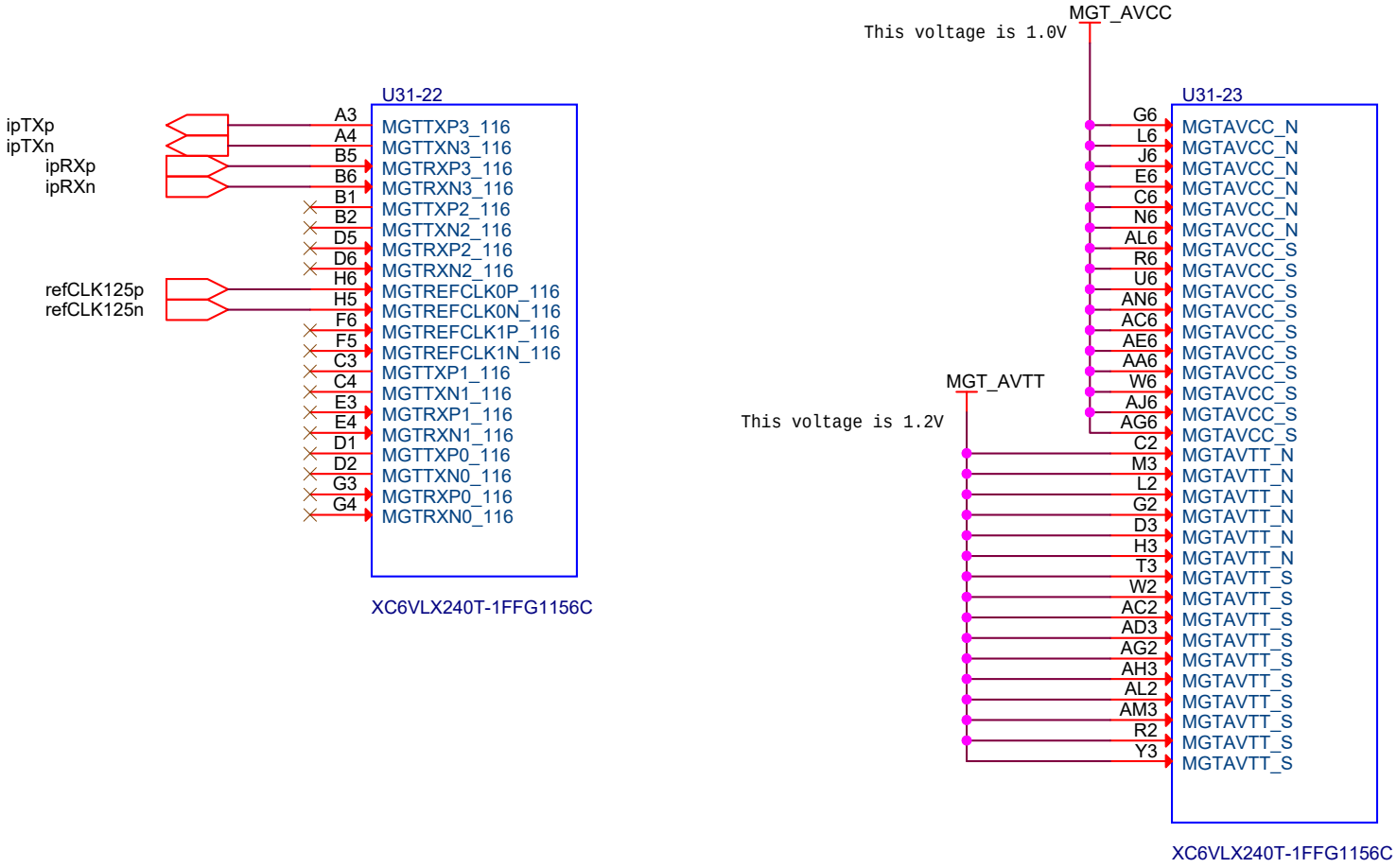
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Designer = Rc Ziegenbein	
Title = 3000-031-51102 ACE-e V6 Image Processing FPGA	
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Designer = Rc Ziegenbein

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This voltage is 1.0V

VCCINT

U31-40

V19	VCCINT_V19
W12	VCCINT_W12
R14	VCCINT_R14
T15	VCCINT_T15
T19	VCCINT_T19
AA20	VCCINT_AA20
N18	VCCINT_N18
P11	VCCINT_P11
P15	VCCINT_P15
P19	VCCINT_P19
V21	VCCINT_V21
T21	VCCINT_T21
U10	VCCINT_U10
U12	VCCINT_U12
U14	VCCINT_U14
U16	VCCINT_U16
U20	VCCINT_U20
R12	VCCINT_R12
T11	VCCINT_T11
V15	VCCINT_V15
R10	VCCINT_R10
R20	VCCINT_R20
T13	VCCINT_T13
R16	VCCINT_R16
AA12	VCCINT_AA12
R18	VCCINT_R18
AA16	VCCINT_AA16
AA14	VCCINT_AA14
W16	VCCINT_W16
AA18	VCCINT_AA18
W20	VCCINT_W20
AB11	VCCINT_AB11
N16	VCCINT_N16
N20	VCCINT_N20
AB13	VCCINT_AB13
P13	VCCINT_P13
AB17	VCCINT_AB17
P17	VCCINT_P17
AB21	VCCINT_AB21
P21	VCCINT_P21
M21	VCCINT_M21
N14	VCCINT_N14
Y11	VCCINT_Y11
AB15	VCCINT_AB15
Y13	VCCINT_Y13
AB19	VCCINT_AB19
Y15	VCCINT_Y15
M19	VCCINT_M19
Y17	VCCINT_Y17
N12	VCCINT_N12
Y19	VCCINT_Y19
Y21	VCCINT_Y21
V11	VCCINT_V11
V13	VCCINT_V13
W14	VCCINT_W14

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This voltage is 2.5V

VCCAUX

U31-41

N22	VCCAUX_N22
P23	VCCAUX_P23
T9	VCCAUX_T9
V9	VCCAUX_V9
W10	VCCAUX_W10
P9	VCCAUX_P9
Y23	VCCAUX_Y23
R22	VCCAUX_R22
AA10	VCCAUX_AA10
U22	VCCAUX_U22
AA22	VCCAUX_AA22
N10	VCCAUX_N10
W22	VCCAUX_W22

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U31-42

U13	GND_U13
R25	GND_R25
F7	GND_F7
Y14	GND_Y14
W5	GND_W5
J7	GND_J7
G29	GND_G29
M4	GND_M4
K10	GND_K10
N5	GND_N5
P16	GND_P16
N21	GND_N21
A1	GND_A1
R9	GND_R9
A2	GND_A2
T22	GND_T22
A5	GND_A5
V10	GND_V10
A6	GND_A6
W23	GND_W23
A7	GND_A7
E15	GND_E15
A17	GND_A17
G1	GND_G1
A27	GND_A27
H26	GND_H26
A34	GND_A34
J33	GND_J33
AA1	GND_AA1
L5	GND_L5
AA2	GND_AA2
M24	GND_M24
AA5	GND_AA5
N13	GND_N13
AA7	GND_AA7
P7	GND_P7
AA9	GND_AA9
P28	GND_P28
AA11	GND_AA11
R17	GND_R17
AA13	GND_AA13
T12	GND_T12
AA15	GND_AA15
U5	GND_U5
AA17	GND_AA17
U29	GND_U29
AA19	GND_AA19
V20	GND_V20
AA21	GND_AA21
W13	GND_W13

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U31-43

AA27	GND_AA27
Y6	GND_Y6
AB3	GND_AB3
Y22	GND_Y22
AB4	GND_AB4
F3	GND_F3
AB7	GND_AB7
F22	GND_F22
AB8	GND_AB8
G7	GND_G7
AB12	GND_AB12
H7	GND_H7
AB14	GND_AB14
J2	GND_J2
AB16	GND_AB16
J13	GND_J13
AB18	GND_AB18
K4	GND_K4
AB20	GND_AB20
K30	GND_K30
AB22	GND_AB22
L17	GND_L17
AB24	GND_AB24
M14	GND_M14
AB34	GND_AB34
N1	GND_N1
AC1	GND_AC1
N9	GND_N9
AC5	GND_AC5
N17	GND_N17
AC7	GND_AC7
P3	GND_P3
AC11	GND_AC11
P12	GND_P12
AC21	GND_AC21
P20	GND_P20
AC31	GND_AC31
R5	GND_R5
AD4	GND_AD4
R13	GND_R13
AD7	GND_AD7
R21	GND_R21
AD18	GND_AD18
T7	GND_T7
AD28	GND_AD28
T16	GND_T16
AE1	GND_AE1
U1	GND_U1
AE2	GND_AE2
U9	GND_U9

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U31-44

AE5	GND_AE5
U19	GND_U19
AE7	GND_AE7
V4	GND_V4
AE15	GND_AE15
V14	GND_V14
AE25	GND_AE25
V26	GND_V26
AF3	GND_AF3
W9	GND_W9
AF4	GND_AF4
W19	GND_W19
AF7	GND_AF7
Y4	GND_Y4
AF12	GND_AF12
Y10	GND_Y10
AF22	GND_AF22
Y18	GND_Y18
AF32	GND_AF32
T8	GND_T8
AG1	GND_AG1
E25	GND_E25
AG5	GND_AG5
F4	GND_F4
AG7	GND_AG7
F12	GND_F12
AG9	GND_AG9
F32	GND_F32
AG19	GND_AG19
G5	GND_G5
AG29	GND_AG29
G19	GND_G19
AH4	GND_AH4
H4	GND_H4
AH7	GND_AH7
H16	GND_H16
AH16	GND_AH16
J1	GND_J1
AH26	GND_AH26
J5	GND_J5
AJ1	GND_AJ1
J8	GND_J8
AJ2	GND_AJ2
J23	GND_J23
AJ5	GND_AJ5
K3	GND_K3
AJ7	GND_AJ7
K7	GND_K7
AJ13	GND_AJ13
K20	GND_K20

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U31-45

AJ23	GND_AJ23
L1	GND_L1
AJ33	GND_AJ33
L7	GND_L7
AK3	GND_AK3
L27	GND_L27
AK4	GND_AK4
M7	GND_M7
AK7	GND_AK7
M20	GND_M20
AK10	GND_AK10
M34	GND_M34
AK20	GND_AK20
N2	GND_N2
AK30	GND_AK30
N7	GND_N7
AL1	GND_AL1
N11	GND_N11
AL5	GND_AL5
N15	GND_N15
AL7	GND_AL7
N19	GND_N19
AL17	GND_AL17
N31	GND_N31
AL27	GND_AL27
P4	GND_P4
AM4	GND_AM4
P10	GND_P10
AM7	GND_AM7
P14	GND_P14
AM8	GND_AM8
P18	GND_P18
AM14	GND_AM14
P22	GND_P22
AM24	GND_AM24
R1	GND_R1
AM34	GND_AM34
R7	GND_R7
AN1	GND_AN1
R11	GND_R11
AN2	GND_AN2
R15	GND_R15
AN5	GND_AN5
R19	GND_R19
AN8	GND_AN8
R23	GND_R23
AN11	GND_AN11
T4	GND_T4
AN21	GND_AN21
T10	GND_T10

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U31-46

AN31	GND_AN31
T14	GND_T14
AP3	GND_AP3
T20	GND_T20
AP4	GND_AP4
T32	GND_T32
AP8	GND_AP8
U2	GND_U2
AP18	GND_AP18
U7	GND_U7
AP28	GND_AP28
U11	GND_U11
AP34	GND_AP34
U15	GND_U15
B3	GND_B3
U21	GND_U21
B4	GND_B4
V3	GND_V3
B7	GND_B7
V7	GND_V7
B14	GND_B14
V12	GND_V12
B24	GND_B24
V16	GND_V16
C1	GND_C1
V22	GND_V22
C5	GND_C5
W1	GND_W1
C7	GND_C7
W7	GND_W7
C11	GND_C11
W11	GND_W11
C21	GND_C21
W15	GND_W15
C31	GND_C31
W21	GND_W21
D4	GND_D4
W33	GND_W33
D7	GND_D7
Y5	GND_Y5
D8	GND_D8
Y7	GND_Y7
D18	GND_D18
Y12	GND_Y12
D28	GND_D28
Y16	GND_Y16
E1	GND_E1
Y20	GND_Y20
E2	GND_E2
Y30	GND_Y30

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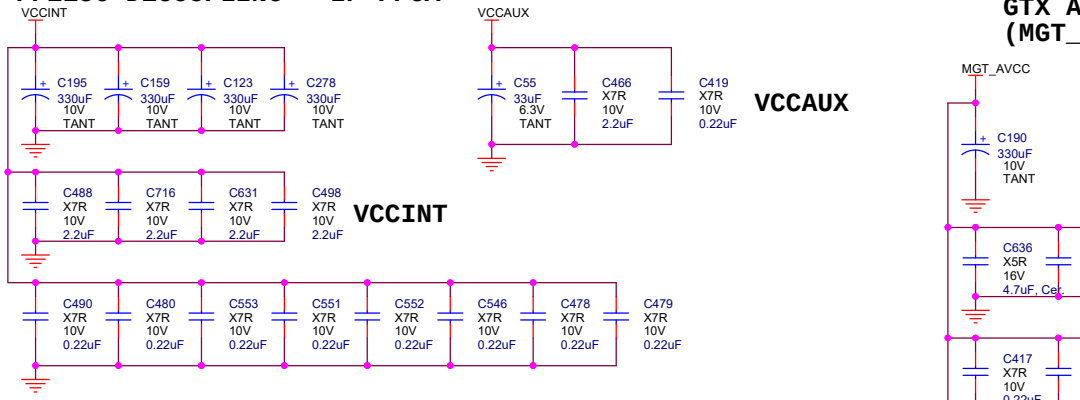
U31-47

E5	GND_E5
E7	GND_E7
G9	GND_G9

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V-6 FF1156 DECOUPLING - IP FPGA

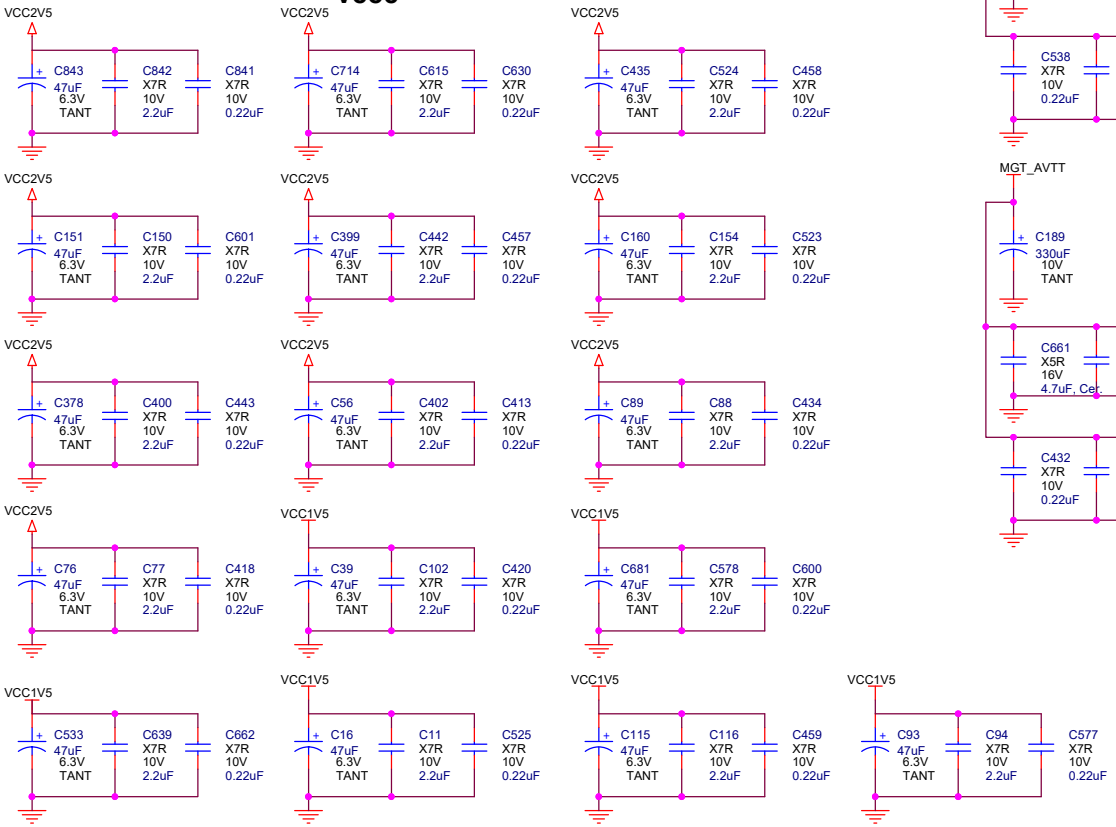


VCCAUX

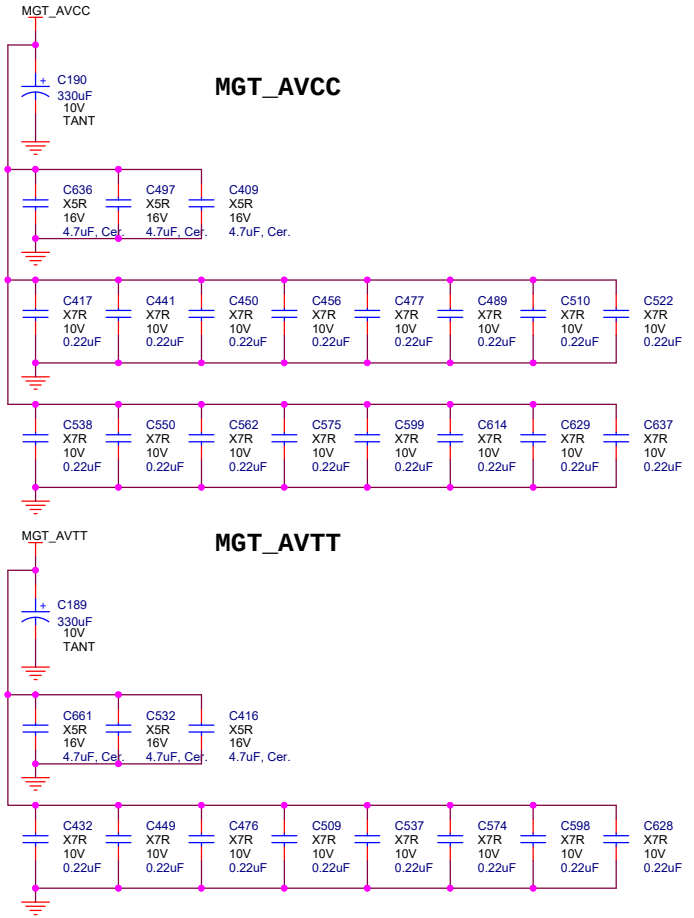
VCCINT

VCCO

One module per bank: 10 banks use VCC2V5, 6 use VCC1V5



GTX ANALOG VOLTAGE SUPPLIES (MGT_AVCC, MGT_AVTT)



MGT_AVCC

MGT_AVTT

PLACEMENT RULES

330uF, 100uF, 47uF, 33uF: Anywhere on board, but as close as possible to FPGA.

4.7uF, 2.2uF: No more than 3" from periphery of FPGA, but as close as possible.

0.22uF: No more than 1" from periphery of FPGA, but as close as possible.

Volcano Corporation 2870 Kilgore Road Rancho Cordova, CA 95670

Designer = M. Shell

Title: 3000-031-51102 ACE-e V6 Image Processing FPGA Bypass

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